Republic of Iraq

Ministry of Higher Education and Scientific research

جمهورية العراق
وزارة التعليم العالي والبحث العلمي

الحقيبة العلمية الدراسية للدراسة الفصلية لمادة الفيروسات

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Target population

طلبة المرحلة الثالثة

Ministry of Higher Education and Scientific Research

College of Health and Medical Techniques

Medical Laboratory Techniques Department

Training package in Medical Virology

The Theoretical Part of Virology lectures-15 Weeks	
Week	**Lecture details**
1	General Properties of Viruses and Structure of Viruses.
2	Classification and Nomenclature of Viruses Atypical Virus-like agents (Prions, Defective viruses, Pseudovirion, and Viriods).
3	Viral Genetic and Viral Replication.
4	Viral Pathogenesis and Transmission.
5	Immunity & Laboratory Diagnosis of Viruses.
6	Herpes virus and Pox virus.
7	Hepatitis virus.
8	Human Immune Deficiency virus.
9	Orthomyxoviruses
10	Paramyxovirus.
11	Enteric viruses (Rota, Polio and Reo viruses).
12	Rabies virus.
13	Coronavirus.
14	Bacteriophages.
15	Oncogenic viruses and Antiviral Drugs &Viral vaccines

2024-2025

Objectives: The study of Virology introduces students to Studying the common steps of the viral reproductive cycle, illustrated with a set of representative viruses, and considering mechanisms by which these viruses can cause disease provides an integrated overview of the biology of these infectious agents.

References:

- Principles of Virology, 5th Edition, 2020, ASM press. By Jane Flint, Vincent R. Racaniello, Glenn F.Rall, Theodora Hatziioannoe, and Ann Marie Skalka.
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- Jawetz, R., J.L. Melnick, and E.A. Adelberg, Review of Medical Microbiology, Twenty-Eighth Edition, 2019.
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LECTURE 1

-General properties of Viruses

-Structure of the viruses.

Definition of viruses

Viruses are the smallest infectious agents and contain only one kind of nucleic acid (RNA or DNA) as their genome. The nucleic acid is encased in a protein shell, which may be surrounded by a lipid-containing membrane (envelop). Viruses are inert in the extracellular environment; they replicate only in living cells, being parasites at the genetic level.

The term virus, which come from the latin word for venom (**poison**).

Because the viruses pass through bacterial filters, therefore the viruses were known as (filterable agents).

Medical Virology :- The science that deal with the study of the medically viruses which infect human.

Virus is a broad general term for any aspect of the infectious agent and includes:-

- The infectious or inactivated virus particle.
- Viral nucleic acid and protein in the infected cell.

Virion:- is the physical particle in the extra-cellular phase which is able to spread to new host cells; complete intact virus particle. The whole virus particle is called (Virion)

General properties of viruses

1- Virus's possession of only one type of nucleic acid, either DNA or RNA, but never both.

2- Viruses are not considered as cell because they do not have a cellular composition. They lack cellular organelles such as nucleus, cytoplasm, mitochondria, ribosome, Golgi apparatus, and endoplasmic reticulum.

3- Viruses are note capable of independent replication, but they replicate only within living (dependent) host cell, therefore they are known as obligate intracellular pathogens.

4- Viruses cannot multiply by binary fission or mitosis, but they multiply by complex process called replication. In which, the viruses produce many copies of their nucleic acid and proteins, and then reassemble into multiple new viruses (progeny viruses).
5- One virus can replicate and produce hundreds of progeny viruses, whereas other organisms, one cell divides to produce only two daughter cells.
6- Virus inside living host cell are active, whereas outside living cells are inert metabolite. Therefore, viruses linked between living and nonliving things.
7- Not all viruses are harmful to human; but the viruses can infect other types of organisms in the nature (such as animals, plants, fungi, bacteria). Therefore, the viruses are not considered to be normal microflora.
8- Viruses cannot grow on inanimate culture media (non-living), but grow in tissue culture (living cells).
9- Viruses cannot be seen by light microscope, but they can seen by electronic microscope.
10- Viruses are unaffected by antibiotic agents but sensitive to antiviral chemotherapy agents and interferon.

Structure of viruses

The main components of viral particle are nucleic acid and proteins:

❖ **Nucleic acid (viral genome):**

- The viruses have central core of nucleic acid, which is either DNA or RNA but not both, therefore the viruses can be divided according to type of nucleic acid into two groups: DNA and RNA viruses.
- The nucleic acid is important part of virus structure because it represent infective particle.
- Viral nucleic acid can be either single stranded (ss) or double strand (ds), linear or circular, segmented or non-segmented genome.
- All viruses contain single copy of genome (haploid), except retroviruses (HIV) have two copies of genome (diploid).

❖ **Capsid (protein coat):**

- The central core is surrounded by protein coat which called capsid.
- The capsid made up number of subunits called **capsomeres**. Each capsomere consisting of one or several proteins known as promoters.

The capsid severs several important functions:

1. The capsid gives shape of virus.
2. Protect viral genetic materials from eternal harmful effects.
3. Mediated attachment of viruses to specific receptor on surface of host cells.
4. Act as antigen that induce neutralizing antibodies and activate cytotoxic T-cell to kill virus-infected cells.

   The unit composed of together nucleic acid and capsid protein is called nucleocapsid (or nucleoprotein, NP)

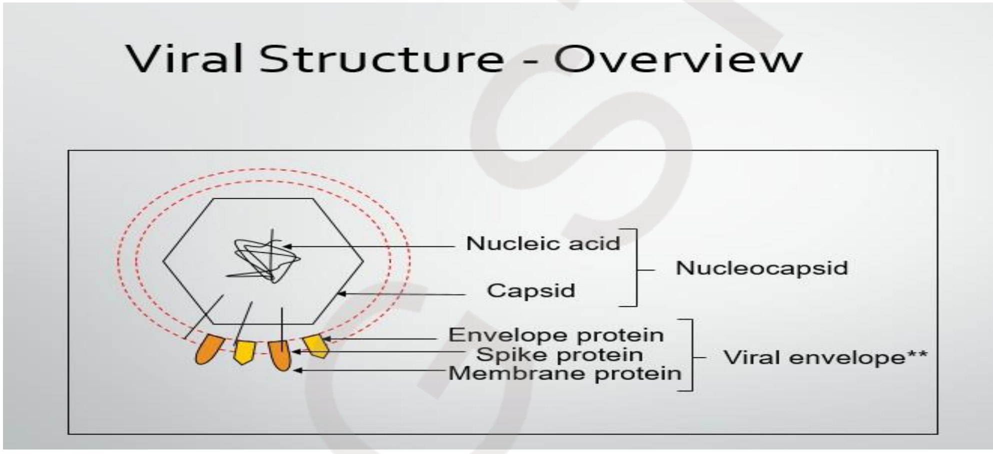


Fig 1. Schematic overview of the structure of animal viruses

** does not exist in all viruses

Other structures:

Envelope : Certain DNA viruses and most RNA viruses are enveloped. The other viruses are non-enveloped (naked). The envelope is consist from lipoproteins which derived from cell membrane of infected cell when virus released by budding from infected cell (except herpes viruses envelope which derived from nuclear membrane of infected cells)

The presence of an envelope confers instability on the virus.

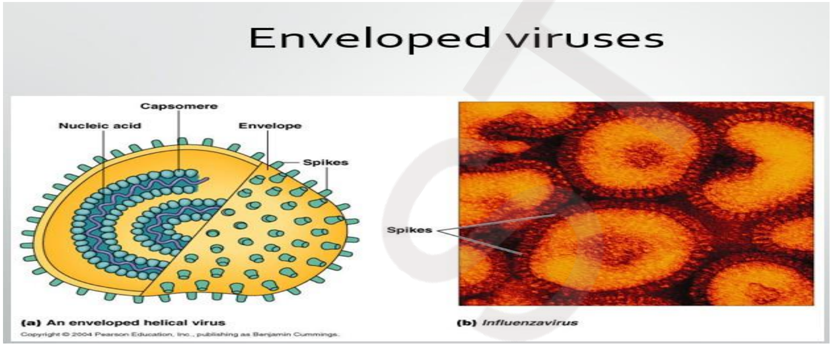
Nucleic acid +capsid + envelope = enveloped Viruses

Spikes : The envelope of certain viruses may be covered with projecting spikes (**glycoproteins**), which called **peplomeres.** The projections may act as viral antigens or may have a role in attachment of virus to cellular receptors.

Matrix protein mediates the interaction between the capsid proteins and enveloped .

Non-structural proteins : Most viral proteins are structural, while other proteins are functional such as viral enzymes, essential for replication of viruses.

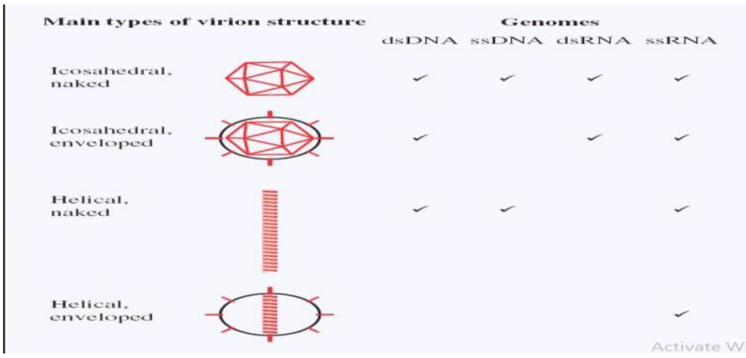
The complete structural unit of entire virus particle is called **virion** the mature virion in some viruses may be consists of only nucleocapsid, whereas in other viruses the virion is more complex, it includes nucleocapsid plus surrounding envelope with or without spikes, the virion is mature infectious particle, by which the virus invade other cells.

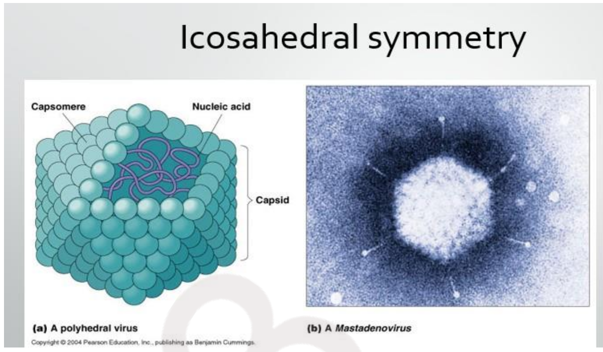


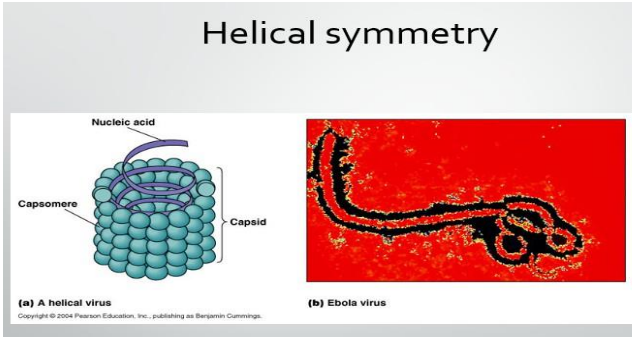
Symmetry types of virus particles:

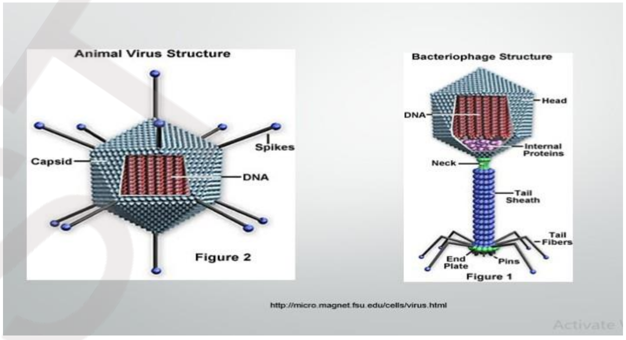
The symmetry depending up on the ways in which the capsomeres are arrangement.

- **Icosahedral symmetry**: is cubic multiple faces(polyhedron), in which the capsomeres are arranged in pattern consisting of multiple triangular faces. Most DNA viruses and some RNA viruses have icosahedral symmetry.
- **Helical symmetry:** in which the capsomeres are arranged in spiral from around nucleic acid that appears rod-shape (tubular shape). The helical symmetry found only in RNA viruses.
- **Complex symmetry**: some viruses don't exhibit icosahedral or helical symmetry but are more complicated in structure, eg; Bacteriophages(viruses infect bacteria) have compex shape consist from head (in icosahedral shape) contain nucleic acid, and tail(in helical shape) has set of fibers which helping in attachment of virus to host cell bacterium.









Reaction to physical and chemical agents:

1. Heat and cold Viral infectivity is generally destroyed by heating at 50-60oC for 30 mint., Viruses can be preserved at -90 oC or -196oC (liquid nitrogens).

2. PH Viruses can be preserved at physiological PH (7.3).

3. Ether susceptibility Ether susceptibility can be used to distinguish viruses that possess an envelope from those that do not.

4. Detergents: Nonionic detergents solubilize lipid constituents of viral membranes. The viral proteins in the envelope are released. Anionic detergents also solubilize viral envelopes; in addition, they disrupt capsids into separated polypeptides.

5. Salts Many viruses can be stabilized by salt in concentrations of 1 mol/L. e.g. $MgCl_2$, $MgSO_4$, Na_2SO_4.

Table (1) : Comparison between viruses and bacteria

Property	Viruses	Bacteria
Size	20-300 nm	1000nm
Genome (type of nucleic acid)	DNA or RNA but not both	DNA and RNA
Cell wall	Envelope present in some viruses	Cell wall
Ribosomes	No Ribosomes	Ribosomes
Multiplication by binary fission	-	+
Sensitivity to antibiotics	-	+
Growth in culture media	Growth only in the living host cell	Grow in culture media

Q1:- Answer True or false ?

1-Viruses contain both types of nucleic acid (DNA and RNA).

2- Spike found on the envelope surface in some viruses.

Q2:- Define the following:

Virion , virus, capsomeres

LECTURE 2-

-Nomenclature and Taxonomy of Viruses

-Atypical Virus-like agents (Prions, Defective viruses, Pseudovirion and Viriods).

Classification of Viruses

Classification of viruses is based on the following characteristics:-

1- Virion morphology, including size, shape, type of symmetry, presence or absence of enveloped.

2. Virus genome properties, including type of nucleic acid (DNA or RNA), size of genome, strandedness (single or double), whether linear or circular, positive or negative sense (polarity), segments (number, size). 3.Physicochemical properties of the virion, including PH stability, thermal stability, and susceptibility to physical and chemical agents especially ether and detergents.

4. Virus protein properties, including number, size and functional activities of structural and non-structural proteins, amino acid sequences, and special functional activities (transcriptase,reverse transcriptase, neuraminidase, fusion activities).

5. Genome organization and replication , including gene order, strategy of replication (patterns of transcription, translation), and cellular sites (accumulation of proteins, virion assembly, virion release).

6. Antigenic properties

7. Biological properties, including natural host range, mode of transmission, vector relationships, pathogenicity, tissue tropisms, and pathology.

Nomenclature of viruses

The viruses are classified into groupings which called families, the family names have the suffix-viridae. Each family, subdivided into genera. The genus names carry the suffix-virus.

The name of viruses are derived from:

1. The name of disease caused by virus (eg: influenza virus, hepatitis virus)
2. Organs or tissues they infect with virus (such as; adenovirus)
3. The geographic locality where the virus was first isolated (such as; West Nile virus).
4. The name of scientists responsible for isolating virus (such as; Epstein-Barr virus)
5. Unique episemiological characteristics of virus (such as: arboviruses; these are arthropod-borne viruses).

Baltimore classification

Viruses were divided into seven groups based on the their nucleic acid and m-RNA production.

1- Double strand DNA (ds-DNA viruses) for example (adenovirus , herpes viruses) .

2- Single strand DNA (ss-DNA viruses) for example (Parvoviruses).

3- ds- RNA viruses(e.g. Reo viruses).

4- (+) ssRNA viruses (+) sense RNA (e.g. Picornaviruses, Togaviruses).

5- (-) ssRNA viruses with (-) sense RNA (e.g. Orthomyxoviruses).

6- ssRNA-Reverse Transcriptase viruses (+) sense RNA with DNA intermediate (e.g. Retroviruses)

7- dsDNA-RT viruses (e.g. Hepadnaviruses). — Universal system of virus taxonomy: Families – on the basis of virion morphology, genome structure and strategies of replication. Virus family names have the suffix – viridae for example Herpesviridae Genera – based on physicochemical or serological differences.

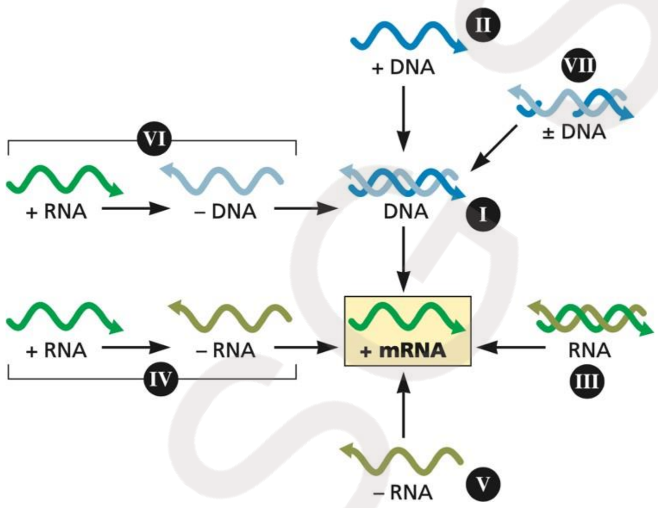


Figure 1.12 The Baltimore classification. The Baltimore classification assigns viruses to seven (I to VII) distinct classes on the basis of the nature and polarity of their genomes. Because all viruses must produce mRNA that can be translated by cellular ribosomes, knowledge of the composition of a viral genome provides insight into the pathways required to produce mRNA, indicated by arrows.

Classification of viruses starts at the level of order and continues as follows, with the taxon suffixes given in italics:

Order (-virales)

Family (-viridae)

Subfamily (-virinae)

Genus (-virus) Species

Species names generally take the form of [Disease] virus. Other classification systems are used to classify viruses into different groups.

-Atypical Virus-like agents (Prions, Defective viruses, Pseudovirion , satellites and Viriods).

satellites: small, single-stranded RNA molecules, which lack genes required for their replication. In the presence of a helper virus, they can replicate, there are two types:

a. satellites viruses: most are associated with plant viruses.
b. Satellites nucleic acids (virusoids).

Defective viruses: are composed of viral nucleic acid and protein, but cannot replicate without co-virus (helper virus) because missing some functions, such as certain adenoviruses and hepatitis D virus) .Defective viruses usually have a mutation or a deletion of part of their genetic material. During the growth of most human viruses, many more defective than infectious virus particles are produced. The ratio of defective to infectious particles can be as high as 100:1 .

Pseudoviruses: the virus particle contains host cell DNA instead of viral DNA within capsid. They are formed during infection of host cell. Pseudoviruses can infect cells but they don’t replicate.

Viroid: consist of only single molecule of circular ssRNA without protein coat or envelope. They replicate and cause several diseases in plant but not in human. Viroids particles that cause potato spindle tuber disease.

<u>Viroids have been found to differ from viruses in six ways:</u>

1. Consists of a single circular RNA molecule of low molecular weight.
2. Exist inside cells, usually inside of nucleoli as particles of RNA without capsids or envelopes.
3. Do not require a helper virus.
4. Viroid RNA does not produce proteins.
5. **Viroid RNA is always copied in the host cell nucleus.**
6. Not apparent in infected tissue without use of special techniques to ID (Identity document) nucleotide sequences in the RNA.

Prion: is infectious particle that is composed only protein. This protein has ability to cause fatal disease. The prion diseases are called spongiform (slowly progressive diseases) include: Scrapie disease in sheep, Mad cow in cattle, and Kuru disease in human.

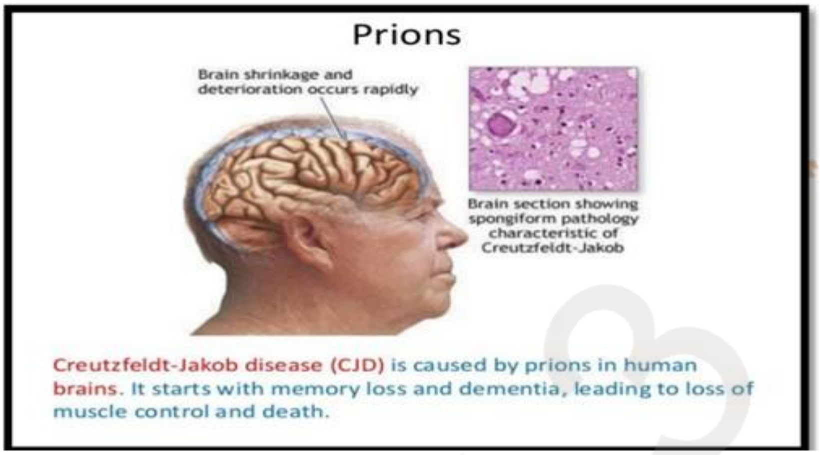
Because neither DNA nor RNA has been detected in prions, they are clearly different from viruses . Furthermore, electron microscopy reveals filament rather than virus particles. Prions are much more resistant to inactivation by ultraviolet light and heat than are viruses. They are remarkably resistant to formaldehyde and nucleases. However, they are inactivated by hypochlorite, NaOH, and autoclaving.

Comparison between prions and conventional viruses

Feature	Prions	Conventional viruses
Nucleic acid	No	Yes
Protein	Yes , encoded by cellular genes	Yes ,encoded by viral genes
Heat inactivation	No	Yes
Appearance	Amyloid- like	Icosahedral
Antibody response	No	Yes
Inflammatory responses	No	Yes

Causes of prion disease

Prion diseases occur when normal prion protein, found predominantly on the surface of neurons, becomes abnormal and clump in the brain, causing brain damage. This abnormal accumulation of protein in the brain can cause memory impairment, personality changes, and difficulties with movement. Experts still don't know a lot about prion diseases, but unfortunately, these disorders are generally fatal.



Q1/ compare between Prion and viroid?

Q2/ Define Pseudovirion , Defective virus?

LECTURE 3 & 4

-Viral Genetic &Viral Replication

-Viral Pathogenesis and Transmission

Viral Genetic

Each virus carries within the protective capsid a nucleic acid-based blueprint for replication of infectious virus particles (virion). Once a virus has invaded a cell, it is able to direct the host cell machinery to synthesize new progeny. The viral genome may be composed of RNA or DNA, single or double stranded. Encoded proteins may be nonstructural, such as nucleic acid polymerases required for replication of genetic material or structural, those proteins necessary for assembly of new infectious virions.

However, all viruses lack the genetic information encoding proteins necessary to generate metabolic energy or protein synthesis. The viral genome RNA or DNA rarely codes for more than few proteins necessary for replication or physical structure. Viruses as a group are the only class of organisms with subspecies that keep RNA as their sole genetic material. Likewise, they are the only group of self-replicating organisms with subspecies that use single-stranded DNA genomic content. Multiple forms of virus genomes are found in virions infecting human cells.

Host cell RNA such as ribosomal RNA can be found in virions, but there is no evidence for a functional role in virus replication. For enveloped viruses, glycoproteins are the major type of protein present on the exterior of the membrane. The existence/presence of a lipid envelope provides an operational method with which to separate viruses into two distinct classes—those that are inactivated by organic solvents.

Multiplication of DNA viruses

As DNA viruses, carry a neutral charge, the DNA is duplicated to mRNA by the action of transcriptase enzyme. mRNA is transferred to the ribosomes for synthesis of (early and late) proteins by translation which are responsible for the formation of a new strand of DNA.

Multiplication of RNA viruses

RNA viruses either carry a –ve or as +ve charge. In positively charged nucleic acid viruses, the nucleic acid itself acts as mRNA and the multiplication goes as mentioned above. While negatively charged viruses, the strand need a transcription for another +ve charge strand by the action of transcriptase enzyme.

Replication of viruses

All of the dynamic events associated with the virus (transcription & replication of genomes) occurs within the living host cell. All DNA viruses multiply inside the host cell nucleus (except poxvirus) and all RNA viruses multiply in the cytoplasm.

Stages of viral replication:

1-Attachment
2-Penetration
3-Uncoating
4- Gene expression and biosynthesis
5-Assembly
6-Release

1-Attachement to host cell

The first stage in viral infection is attachment of virus to specific receptor on the surface of host cell:

A- Receptor molecules **differ for different viruses**. For example, HIV attaches to CD4 receptor on helper T-cell and Rabies virus binds to acetylcholine receptor.

B- The attachment of virus **determines the organ specificity** such as hepatitis virus infect liver, influenza virus infect respiratory tract, and so on.

C- The specificity of attachment **determines the host rang** of viruses. Some viruses have the narrow range, whereas others have abroad ranged.

2- Penetration

Following attachment, virions can enter cells by one of the following ways:

A- Translocation of virion across plasma membrane.
B- Endocytosis: in which the virus is accumulated inside cell.
C- Fusion with Plasma Membrane : The virus fuses directly with the plasma membrane of the cell, and enter into host cell.

3- Uncoating

Uncoating occurs concomitantly with or shortly after penetration, **Uncoating is removing the capsid proteins**. Uncoating may be occurring **in cytoplasm** or **in nucleus**. A low pH within the vesicle and presence of cellular enzymes which lead to dissolve the proteins of capsid , then result in uncoating and release of viral nucleic acid into infected host cell. The viral nucleic acid may remain in cytoplasm or migrate to nucleus.

4- Gene expression and biosynthesis:

Virus cannot replicate by binary fission or mitosis, but they replicate by complex process. When the viral genome released inside living host cell, the virus is control on host cell biosynthesis, inhibition of macromolecules synthesis and use the energy of host cell in synthesis of viral macromolecules.

<u>The gene expression involves</u>

A- **Replication of viral genome (synthesis of viral nucleic acids):** The **DNA** viruses replicate in nucleus (except pox viruses in cytoplasm), whereas the **RNA** viruses are replicate in the cytoplasm (except retro viruses and influenza virus in nucleus).

B- **Transcription of viral mRNA**: synthesis of mRNA in viruses in various pathways, transfer of genetic information from parental genome to mRNA is called transcription.

C- **Translation of mRNA (synthesis of viral proteins):** Once the mRNA of either DNA viruses or RNA viruses is synthesized, and it translated by ribosome of host cell into viral protein.

5- Assembly of Viruses

The virus produces many copies of their nucleic acid and proteins. The newly synthesized viral genome and structural proteins are assembling to form many progeny viruses.

The packaging of viral nucleic acid into capsid is accruing either in cytoplasm or in nucleus of infected cell.

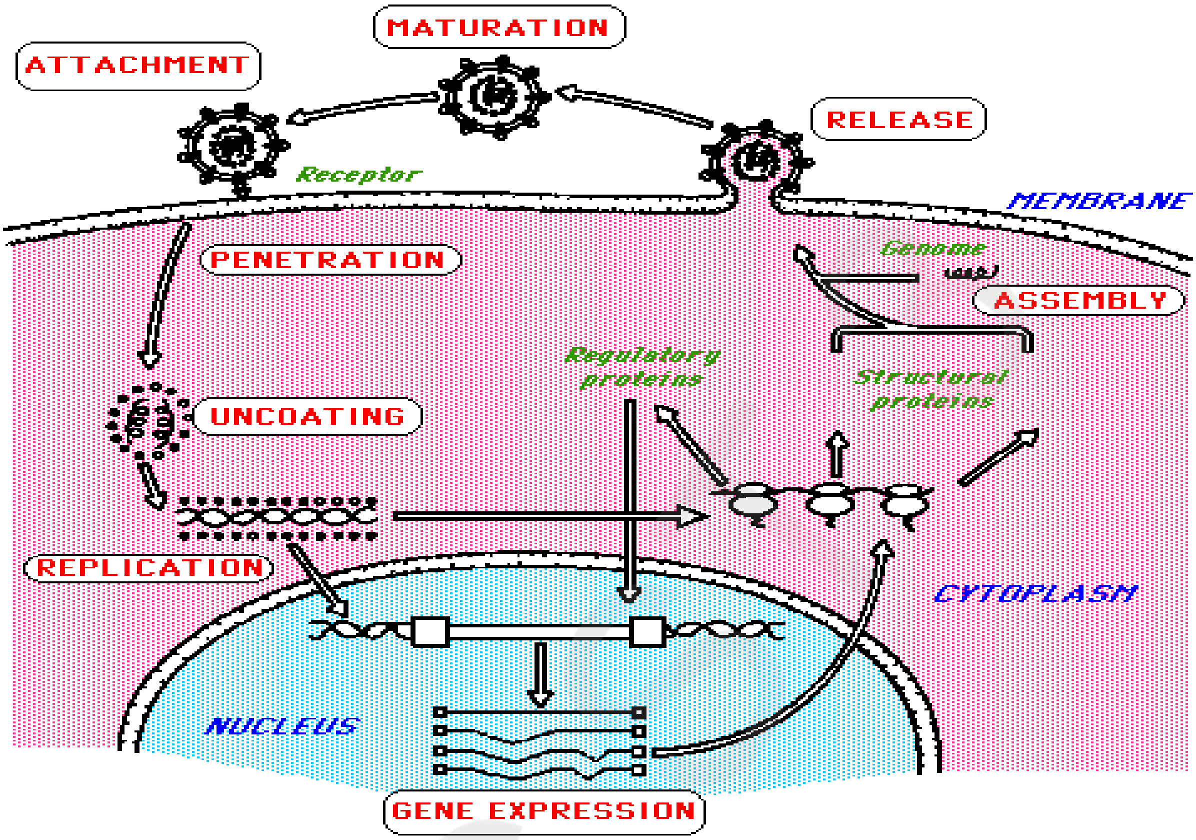
6- Release of virus

The virus mature particles are released from the infected cell by one of two processes:

A- Lysis of infected cell.

B- Budding (without lysis) through the outer cell membrane.

Some viruses are enveloped; they acquired their envelopes from cell membrane during releasing, while other enveloped viruses acquire their envelope from nuclear membrane of infected cell.



Viral Pathogenesis and Transmission

Viral Pathogenesis

Viral pathogenesis: is the process by which a viral infection leads to disease in the host.

➢ **Pathogenic mechanisms:**

1- Implantation of virus at the portal of entry.
2- Local replication.
3- Spread to target organs (disease sites).
4- Spread to sites of shedding of virus into the environment.

➢ **Many factors affect pathogenic mechanisms:**

1- **Accessibility of virus to tissue**
2- **Cell susceptibility to virus multiplication**
3- **Virus susceptibility to host defenses**
4- **The virulence characteristics of the infecting virus.**

➢ Outcome of Viral Infection

❖ Acute Infection

- ✓ Recovery with no residue effects
- ✓ Recovery with residue effects e.g. acute viral encephalitis leading to neurological sequelae.
- ✓ Death
- ✓ Proceed to chronic infection

❖ Chronic Infection

1- Silent subclinical infection for life e.g. CMV
2- A long silent period before disease e.g. HIV
3- Reactivation to cause acute disease e.g. herpes.
4- Chronic disease with relapses and exacerbations e.g. HBV, HCV.
5- Cancers e.g. (Human herpes-8) HHV-8

➢ Factors included in Viral Pathogenesis

For pathogenic virus, there are a number of critical stages in replication which determines the nature of disease they produce which included;

1-Entry into the Host
2-Course of Infection (Primary Replication ,Secondary Replication, Systemic Spread, spread throughout the host)
3-Cell/Tissue Tropism
4-Cell/Tissue Damage
5-Host Immune Response
6-Virus Clearance or Persistence

1- Entry into the host

The first stage in any virus infection. In the case of pathogenic infections, the site of entry can influence the disease symptoms produced. Infection can occur via:

A- Skin
B- Respiratory tract
C- Gastrointestinal tract
D- Genitourinary tract
E-Conjunctiva and other mucous membranes

2-Course of Viral Infection

❖ Primary Replication

After entry to potential host, the virus must initiate an infection by entering a susceptible cell. This frequently determines whether the infection will remain localized at the site of entry or spread to become systemic infection.

Localized infections

Virus	primary replication
Rhinoviruses	upper respiratory tract
Papillomaviruses	Epidermis
Rotaviruses	Intestinal epithelium

❖ Secondary replication

Occurs in systemic infections when a virus reaches other tissues in which it is capable of replication, e.g. poliovirus (gut epithelium- nervous in brain &spinal cord). If the virus can be prevented from reaching tissues where secondary replication can occur, generally no disease results.

Systemic Infections

Virus	primary replication	secondary replication
Enteroviruses	Intestinal epithelium	Lymphoid tissues, C.N.S
Herpesviruses	Oropharynx or G.U. tract	Lymphoid cells, C.N.S

❖ Spread throughout the host

Apart from direct cell-cell contact, there are 2 main mechanisms for spread throughout the host:

- **Via the bloodstream**
- **Via nervous system**

The virus may get into the bloodstream by direct inoculation- e.g. Arthropod vectors, blood transfusion, or I.V drug abuse.

The virus may travel free in the plasma (Togaviruses, Enteroviruses) or in association with red cells (Orbiviruses) platelets (Herpes simplex virus). Spread to the nervous system is preceded by primary viremiaa.

3- Cell/ Tissue tropism

Tropism- is the ability of a virus to replicate in particular cells or tissues- is controlled partly by the route of infection but largely by the interaction of a virus attachment protein (V.A.P) with a specific receptor molecule on the surface of a cell and has a considerable effect on pathogenesis.

4-Host immune response

Has a major impact on the outcome of an infection. In the most cases the virus is cleared completely from the body and results in complete recovery. In other infections, the immune response is unable to clear the virus completely and the virus persists. In general, cellular immunity plays the major role in clearing virus infection whereas humoral immunity protects against reinfection.

5- Cell /Tissue damage

Viruses may replicate widely throughout the body without any disease symptoms if they do not cause significant cell damage or death. Retroviruses do not generally cause cell death, being released from the cell by budding rather than by cell lysis and cause persistent infections, even being passed vertically to offspring if they infect the germline.

6- Viral Clearance or Persistence

The majority of viral infections are cleared but certain viruses may cause persistent infections. There are 2 types of chronic persistent infections :

1-True Latency -the virus remains completely latent following primary infection e.g. Herpes simplex virus .
2- Persistence - the virus replicates continuously in the body at a very low level e.g. (HIV).

Virulence and cytopathogenicity

Viral strains that kill target cells and cause disease are called virulent viruses, but other strains that have mutated and lost their ability to cause cytopathic effect (CPE) and disease are termed as nonvirulent or attenuated strains. Some attenuated strains can be used as live vaccines. Examples are MMR (measles, mumps, rubella).

Virulence: is the relative ability of a virus to cause disease, virulence can be measured as the degree of pathogenicity between closely related viruses to cause disease.

Cytopathogenicity: is the ability of a virus to cause degenerative changes in cells or cell death.

Virulence and cytopathogenicity depend on the nature of viruses and the characteristics of cells such as permissive and non-permissive cells.

a- A permissive cell: permits production of progeny virus particles and \ or viral transformation.
b- A non-permissive cell does not allow virus replication, but it may permit the transformation of the cell.

- Replication of the virus results in alterations of cellular morphology and function, when a lytic virus infects a permissive cell, lots of daughter viruses are produced and this is followed by lysis of the infected cells, called **cytopathic effects (CPE**) of the virus and that changes of the cell followed by cell death.

The features of CPE are morphologic changes of the cell organelles including:

1- Nucleus (inclusion bodies, thickening of the nucleus, swelling, nucleolar changes, margination of chromatin)
2- Cytoplasm (inclusion bodies, vacuoles).
3- Membranes (cells round up, loss of adherence, cell fusion {syncytial})
4- Cellular lysis (disintegration).

Transmission of human viruses

Infection can be **direct**, for instance, respiratory spread of influenza virus, or **indirect**, for example, arboviruses (west nile virus, yellow fever virus, dengue virus) transmission involving a mosquito vector.

Viruses are transmitted via 3 ways:

1- Horizontal (common route of transmission : person to person)

2- Vertical (mother –to-child transmission) routes can occur in utero, during delivery (via birth canal), and through breast-feeding. Some viruses are transmitted through sexual routes.
3- Zoonotic (animal-to-human) transmission of viral infections can occur from the bite of animals (eg, rabies) or insects (eg, dengue, yellow fever, west Nile) or from inhalation of animal excreta (eg, hantavirus, arenavirus).

Routes of transmission

1- Direct Contact Transmission

Direct contact transmission occurs through direct body contact with the tissues or fluids of an infected individual. Physical transfer and entry of microorganisms occurs through mucous membranes (e.g., eyes, mouth), open wounds, or abraded skin. Direct inoculation can occur from bites or scratches.

2- Fomite Transmission

Fomite transmission involves inanimate objects contaminated by an infected individual that then come in contact with a susceptible animal or human. Fomites can include a wide variety of objects such as exam tables, cages, medical equipment, environmental surfaces, and clothing. Disease examples include canine parvovirus and feline calicivirus infections.

3- Aerosol (Airborne) Transmission

Aerosol transmission encompasses the transfer of pathogens via very small particles or droplet nuclei. Aerosol particles may be inhaled by a susceptible host or deposited onto mucous membranes or environmental surfaces. This can occur from breathing, coughing, sneezing, or vocalization of an infected individual, but also during certain medical procedures (e.g., bronchoscopy, dentistry, inhalation anesthesia). Very small particles may remain suspended in the air for extended periods and be disseminated by air currents in a room or through a facility. However, most pathogens pertinent to companion animal veterinary medicine do not survive in the environment for extended periods or do not travel great distances due to size and as a result require close proximity or contact for disease transmission. Examples of common aerosolized pathogens include canine influenza, and canine distemper virus.

4- Oral (Ingestion) Transmission

The ingestion of pathogenic organisms can occur from contaminated food or water as well as by licking or chewing on contaminated objects or surfaces. Environmental contamination is most commonly due to exudates, feces, urine, or saliva. Examples of diseases acquired via oral transmission include feline panleukopenia.

5- Vector-Borne Transmission

Vectors are living organisms that can transfer pathogenic microorganisms to other animals or locations and include arthropod vectors (e.g., mosquitoes, fleas, ticks) and rodents or other vermin. Vector-borne transmission can be an important route of transmission in climates where these pests exist year-round and may be brought into the practice by an infested patient. Examples of vectorborne disease is plague.

6- Sexually transmitted

Sexually transmitted of viruses are more common.

Important terms

Viral Disease: is the result of complex interactions between the virus and susceptible host.

Infectivity: is the frequency with which an infection is transmitted when there is contact between a virus and a susceptible host, and represents the ability of the virus to infect an individual. Measures of infectivity are generally expressed as attack rates (number of persons infected after exposure/ the number of susceptible persons).

Prevalence: is the total number of cases of disease in a population during a defined time period (number of new and old cases of disease \ population at-risk), and represents the burden of disease in a population.

Incubation period: is the time between exposure to the organism and appearance of the first symptoms of the disease (after virus entry into the host, viruses have variable incubation periods).

Epidemiology: deals with distribution and determinants of disease in human populations.

- **Endemic** (disease present at fairly low, but constant level).
- **Epidemic** (infection greater than normally occurs in the population)
- **Pandemic** (infections that are spread worldwide involving a novel virus and person-to-person spread).

<u>Questions</u>

1- -------------- is defined as the time from the onset of infection to the appearance of virus extracellularly.

2- Enumerate the replication steps of virus?
3- Mention the types of viral release ways from the host cell infection ?

What is the pathogenesis of virus ?
Write the important routs for transmission of viruses?

What are you know about the following?
- Incubation period
- vertical transmission
- horizontal transmission
- Animal-to-human transmission

Q/ give definition to:
Virulence, pathogenicity, Cytopathogenicity, Tropism, infectivity, epidemiology, Pandemic.

LECTURE- 5

-Immunity & Laboratory Diagnosis of Viruses

A- innate immune-response (non-specific response)

1. Interferon: Alpha and beta interferon are group of proteins produced by human cells after viral infection(have antiviral effects).

- Inhibit the growth of viruses by blocking the synthesis of viral proteins
- Prevents further spread of viruses
- Prevents uninfected cells from kill by NK

Interferon have two main mechanisms

- Ribonuclease that degrades mRNA
- Protein kinase that inhibits protein synthesis

2. Natural killer cells (NK) / important part of the innate immunity against virus infected cells , called natural killer cells because they are active without thenecessity of being exposed to the virus previously andthey are not specific for any virus

3. Macrophages / phagocytosis virus and virus infected cells , production of antiviral molecules such as INF

4. Fever / elevated body temperature may play a role in host defenses , but it is important is uncertain

B-Adaptive immune response (Specific response)

1- Antiviral antibodies / such as IgA confers protection against viruses that enter through the respiratory and gastrointestinal mucosa, IgM and IgG protect against viruses that enter or are spread through the blood

2- Cytotoxic T- lymphocytes / CD8 – positive Tcells recognize viral antigen in association with class I MHC proteins , they kill virus infected cells by two methods

- By releasing prteolytic enzymes called granzymes into the infected cell , which degrade the cell contents
- Activating programmed cell death (apoptosis)

Diagnosis of Viral Infections

Sampling

A wide variety of samples can be used for virological testing. The type of sample sent to the laboratory often depends on the type of viral infection being diagnosed and the test required. Proper sampling technique is essential to avoid potential pre-analytical errors. For example, different types of samples must be collected in appropriate tubes to maintain the integrity of the sample and stored at appropriate temperatures (usually 4°C) to preserve the virus and prevent bacterial or fungal growth. Sometimes multiple sites may also be sampled.

Types of samples include:

- Blood
- Skin
- Sputum, gargles and bronchial washings
- Urine
- Semen
- Faeces
- Cerebrospinal fluid
- Tissues (biopsies or post-mortem)
- Dried blood spotsVirus isolation

Viruses are often isolated from a patient's first sample. This allows the virus sample to be grown into larger quantities and allows a larger number of tests to be run on them.

This is particularly important for samples that contain new or rare viruses for which diagnostic tests are not yet developed.

Many viruses can be grown in cell culture in the lab. Other viruses may require alternative methods for growth such as the inoculation of embryonated chicken eggs (e.g. avian influenza viruses) or the intracranial inoculation of the virus using newborn mice (e.g. lyssa viruses)

Virus diagnosis methods

A-Nucleic acid based methods

Molecular techniques are the most specific and sensitive diagnostic tests. They are capable of detecting either the whole viral genome or parts of the viral genome. In the past nucleic acid tests have mainly been used as a secondary test to confirm positive serological results. Molecular methods do not rely on the presence of a live virus like virus isolation procedures.

1- Polymerase chain reaction

Detection of viral RNA and DNA genomes can be performed using polymerase chain reaction. This technique makes many copies of the virus genome using virus-specific probes. Variations of PCR such as nested **reverse transcriptase PCR** and **real time PCR** can also be used to determine viral loads in patient serum. This is often used to monitor treatment success in HIV cases.

2- Sequencing

Sequencing is the only diagnostic method that will provide the full sequence of a virus genome. Hence, it provides the most information about very small differences between two viruses that would look the same using other diagnostic tests. For example, sequencing is useful when specific mutations in the patient are tested in order to determine antiviral therapy and susceptibility to infection.

However, as the tests are getting cheaper, faster, and more automated, sequencing will likely become the primary diagnostic tool in the future.

B-Microscopy based methods

Immunofluorescence or immunoperoxidase assays are commonly used to detect whether a virus is present in a tissue sample. These tests are based on the principle that if the tissue is infected with a virus, **an antibody specific to that virus will be able to bind to it. To do this,** antibodies that are specific to different types of viruses are mixed with the tissue sample. After the tissue is exposed to a specific wavelength of light or a chemical that allows the antibody to be visualized.

Electron microscopy is a method that can take a picture of a whole virus and can reveal its shape and structure. It is not typically used as a routine diagnostic test as it requires a highly specialized type of sample preparation, microscope and technical expertise.

C-Antibody detection

A person who has recently been infected by a virus will produce antibodies in their bloodstream that specifically recognize that virus. This is called humoral immunity. Two types of antibodies are important. The first called IgM is highly effective at neutralizing viruses but is only produced by the cells of the immune system for a few weeks. The second, called, IgG is produced indefinitely. Therefore, the presence of IgM in the blood of the host is used to test for acute infection, whereas IgG indicates an infection sometime in the past. Both types of antibodies are measured when tests for immunity are carried out.

1-ELISA assay

Antibody testing has become widely available. It can be done for individual viruses (e.g. using an ELISA assay) but in automated panels that can screen for many viruses at once are becoming increasingly common.

2-Hemagglutination assay

Some viruses attach to molecules present on the surface of red blood cells, for example, influenza virus. A consequence of this is that at certain concentrations a viral suspension may bind together (agglutinate) the red blood cells thus preventing them from settling out of suspension.

<u>Questions</u>

1-Mention Laboratory diagnosis of viruses?

2-Enumerate the innate immune-response (non-specific response)? Mention two mechanism of interferon?

LECTURE-6
Herpes virus and Pox virus

Introduction

Nearly everyone will be affected by some type of herpes virus at some time in his or her life. Herpes simplex virus belongs to the family of herpesviridae which consists of more than 80. distinct types of viruses that are found in nearly every kind of animals such as fishes, frogs, birds ,cats, dogs, mice, snakes, lizards, monkeys, cows, horses, humans, and more.

Classification

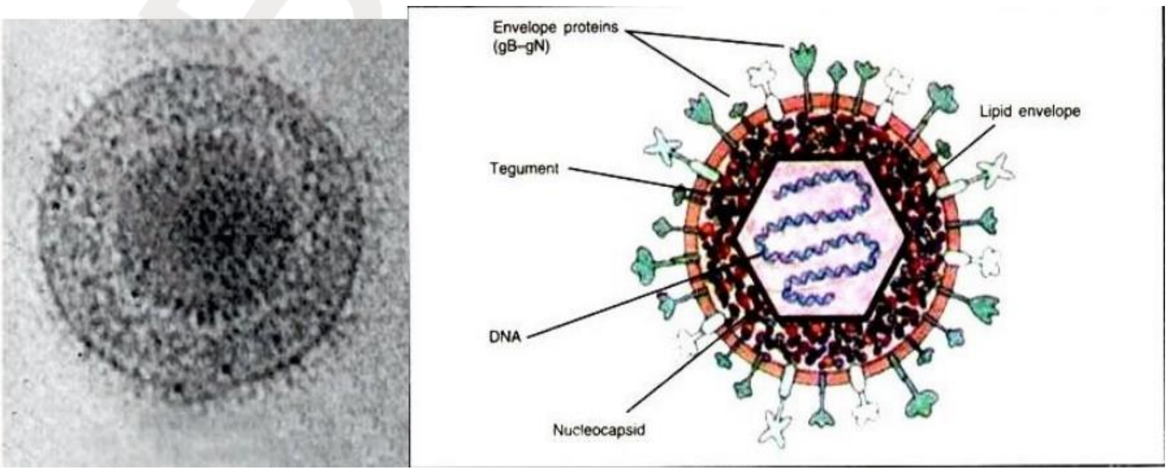
The classification of herpes viruses is complex! All herpes viruses share a common morphology and have genomes of linear, doublestranded DNA (dsDNA) butThe family Herpesviridae is further subdivided into three subfamilies: Alphaherpesvirinae, Betaherpesvirinae and Gammaherpesvirinae, and the subfamilies are subdivided into genera.

TABLE 57-1 Classification of human herpesviruses

Subfamily	Scientific name	Common name	Site of latency
Alphaherpesvirinae	Human herpesvirus 1	Herpes simplex virus type 1	Neurons
	Human herpesvirus 2	Herpes simplex virus type 2	Neurons
	Human herpesvirus 3	Varicella zoster virus	Neurons
Gammaherpesvirinae	Human herpesvirus 4	Epstein–Barr virus	Lymphoid tissues
	Human herpesvirus 8	Kaposi's sarcoma-related virus	–
Betaherpesvirinae	Human herpesvirus 5	Cytomegalovirus	Monocytes and lymphocytes in secretory glands
	Human herpesvirus 6	Human B cell lymphotrophic virus	Lymphoid tissue
	Human herpesvirus 7	RK virus	Lymphoid tissue

Virion Properties

Herpes virus virions are enveloped and include a core, capsid, and tegument (a layer of globular material surrounding the capsid which is enclosed by a typical lipoprotein envelope with numerous small glycoprotein spikes). The nucleocapsid is spherical icosahedral . The core consists of the viral genome packaged as a single, linear dsDNA molecule within the protein capsid. Because of the variable size of the envelope, virions can range in diameter from 120 to 250 nm.



Types of human herpes viruses and their Characteristics

1- Herpes simplex virus type 1 (HSV-1)/

Human herpes virus-1 Herpes simplex virus type 1 causes mainly oral and ocular manifestations. Fever blister or cold sores of the face, mouth, and lips are the most common symptoms of HSV-1 outbreaks.

Besides causing cold sores and possibly spreading to the genital region, HSV-1 has now been linked with the development of serious neurological diseases such as Alzheimer's disease. One can contract HSV-1 through direct contact with sites of viral shedding or mucocutaneous fluids . Some of the infections caused by HSV-1 are pharyngitis, tonsillitis, gingivostomatitis, stromal keratitis, retinitis, encephalitis, pneumonitis, esophagitis, and hepatitis. The recurrences of HSV-1 can be triggered either by internal factors (fever, illness, menstruation, gastrointestinal and respiratory infections, diabetes, hyperthyroidism, fatigue and factors that depress the immune system); or external factors such as exposure to wind, burn, ultraviolet radiation and emotional stress .

2- Herpes simplex virus type 2 (HSV-2)/

Human herpes virus-2 Human herpes virus-2 is the typical cause of genital herpes and classified as a sexually transmitted disease. The adolescents and adults are mainly affected by HSV-2. Excretion of the virus from herpetic lesions mainly lips and genitals of asymptomatic people are regarded as the most important source of the virus. The genital herpes is characterized by painful ulcerated lesions and sores on the genitalia, which increases the risk of contracting HIV by two or three times, as it is believed that 40-60% of HIV patients showed prior infection with HSV-2. Recurrent genital herpes is mainly caused by HSV- 2 through direct contact with sites of viral shedding after reactivation of the virus from its latency in sacral ganglia during sexual intercourse; transferred from mother to child through contact with an infected birth canal, infected caregiver or through ingestion of infected maternal secretions, known as intrauterine infections during child birth. The neonatal infections by HSV-2 pose serious threats in the skin, eye, mouth, herpes encephalitis, and meningitis and dispersed multi-organ infections as they often lead to fatal diseases

3- Herpes zoster virus (HZV)/ Varicella zoster virus (VZV)/

Human herpes virus-3 Chickenpox is the results of first-time infection by HZV and the virus causes shingles when it reappears in a person's life. Rarely, HZV may cause a life threatening due to the infection on central nervous system.

4- Epstein-Barr virus (EBV)/ Human herpes virus-4

Epstein-Barr virus is the major cause of infectious mononucleosis (kissing disease), chronic fatigue syndrome, and disorders of the immune system and linked with lupus, lymphomas, and other cancers. EBV is now considered to be quite damaging and causing genetic mutations in the body

5- Cytomegalovirus (CMV)/ Human herpes virus-5

Cytomegalovirus can cause mononucleosis, hepatitis and can be transmitted through sexual contact . The occurrence of CMV is strongly correlated with asymptomatic vascular diseases such as heart, coronary artery and atherosclerosis. CMV infects about 60% of adults, among homosexual men and is associated with HIV. CMV infection during pregnancy carries a high risk of intrauterine transmission which may result in growth retardation, jaundice, and central nervous system abnormalities. Those who are asymptomatic at birth may develop hearing defects or learning disabilities later in life.

6- Human herpes viruses - Types 6, 7 and 8 (HHV- 6, HHV-7 and HHV-8)
All human herpes viruses are associated with disorders of the immune system.

Viral replication

Herpes simplex virus (HSV) grows very rapidly in infected cells, requiring only 8–16 hours for completion. The virus infects most types of cells in human hosts and usually causes lytic infections of the fibroblasts and epithelial cells. After entry into the cell, the virion is uncoated, genome is released, and the genome DNA enters into the nucleus. The mRNA is transcribed by host cell RNA polymerase and then translated into early nonstructural proteins. Subsequently, the viral DNA polymerase replicates the genome DNA, during which synthesis of early proteins is stopped but synthesis of late structural proteins begins. These late proteins are transported to the nucleus where assembly of virion occurs. The virion acquires its envelope by budding through the nuclear membrane. It also causes latent infections of neurons by the presence of multiple copies of HSV-1 DNA in the cytoplasm of infected neurons.

Diagnosis

Clinical examination of persons with no previous history of lesions and contact with an individual with known HSV-1 infection. The appearance and distribution of sores in these individuals typically presents as multiple, round, superficial oral ulcers accompanied by acute gingivitis, Also the diagnostic methods for detection of herpesvirus:-

- Virus-Specific Polymerase Chain Reaction (PCR).
- Electron Microscopy Of Vesicular Fluid Or Scrapings.
- Immunofluorescence Staining Of Smears Or Tissue Sections.
- Virus Isolation And Characterization Provide A Definitive Diagnosis.
- Specific enzyme immunoassays for antibody detection (serology).

Immunity, Prevention and Control

Control strategies are directed at management practices and vaccination. Bovine herpes virus 1 vaccines are used extensively, alone or in combination as multiple virus formulations , Inactivated and live-attenuated vaccines are available and recombinant DNA vaccines Although they do not prevent infection, vaccines significantly reduce the incidence and severity of disease. Prevention of the HSV 1 infection can be achieved by avoiding the physical contact, kissing when the lesions are present, touching or using the patient's eating or drinking utensils. Even though antiviral treatments can reduce the frequency, and severity of outbreaks, there is no treatment that can eradicate herpes simplex virus-1.

However, asymptomatic carriers of the HSV-2 are still contagious. Some asymptomatic individuals are unaware of their infection, they are considered at high risk for spreading HSV.

POXVIRUSES

<u>General features and diseases caused by Poxviruses: -</u>

-Poxviruses are Very large, brick-shaped viruses about 300 x 200 nm (the size

of small bacteria).
-They have a complex internal structure - a large double-stranded DNA genome is enclosed within a "core" that is flanked by 2 "lateral bodies".
-The surface of the virus particle is covered with filamentous protein components, so that the particles have the appearance of a "**ball of knitting wool**".
-The entire particle is enclosed in an envelope derived from the host cell membranes.
-Most poxviruses are host-species specific, but vaccinia is a remarkable exception.

- True pox viruses are antigenically rather similar, so that infection by one elicits immune protection against the others. **Four orthopoxviruse**s cause infection in humans: **variola, vaccinia, cowpox, and monkeypox.** Variola virus infects only humans in nature, although primates and other animals have been infected in an experimental setting. Vaccinia, cowpox, and monkeypox viruses can infect both humans and other animals in nature.

- **Chickenpox** is not a true poxvirus and is caused by the herpesvirus varicella zoster.
-The life cycle of poxviruses is complicated by having multiple infectious forms, with differing mechanisms of cell entry. Poxviruses are unique among human DNA viruses in that they replicate in the cytoplasm of the cell rather than in the nucleus. To replicate, poxviruses produce a variety of specialized proteins not produced by other DNA viruses, the most important of which is a viral- associated DNA-dependent RNA polymerase. Both enveloped and unenveloped virions are infectious. The viral envelope is made of modified Golgi membranes containing viral-specific polypeptides, including hemagglutinin. Infection with

either variola major virus or variola minor virus confers immunity against the other.

Laboratory Diagnosis of Poxviruses: -

- Microscopically, poxviruses produce characteristic cytoplasmic inclusion bodies, the most important of which are known as Guarnieri bodies, and are the sites of viral replication. Guarnieri bodies are readily identified in skin biopsies stained with hematoxylin and eosin, and appear as pink blobs. They are found in virtually all poxvirus infections but the absence of Guarnieri bodies could not be used to rule out smallpox.

-The diagnosis of an orthopoxvirus infection can also be made rapidly by electron microscopic examination of pustular fluid or scabs. All orthopoxviruses exhibit identical brick-shaped virions by electron microscopy. If particles with the characteristic morphology of herpesviruses are seen this will eliminate smallpox and other orthopoxvirus infections.

-Definitive laboratory identification of variola virus involved growing the virus on chorioallantoic membrane (part of a chicken embryo) and examining the resulting pock lesions under defined temperature conditions.

- Strains were characterized by polymerase chain reaction (PCR).

-Serologic tests and enzyme linked immunosorbent assays (ELISA), which measured variola virus-specific immunoglobulin and antigen were also developed to assist in the diagnosis of infection.[68]

Chickenpox was commonly confused with smallpox in the immediate post-eradication era. A variety of laboratory methods were available for detecting chickenpox in the evaluation of suspected smallpox cases.

-Some pox viruses can be isolated by cell-culture.

Smallpox virus

-belongs to the genus Orthopoxvirus. The last naturally occurring case was diagnosed in October 1977, and the World Health Organization (WHO) certified the global eradication of the disease in 1980, making it the only human disease to be eradicated.

-There are two forms of the smallpox, variola major is the severe and most common form, with a more extensive rash and higher fever. Variola minor is a less common presentation, causing less severe disease, typically discrete smallpox, with historical death rates of 1% or less.

- **Subclinical (asymptomatic**) infections with variola virus were noted but were not common. In addition, a form called variola sine eruption (smallpox without rash) was seen generally in vaccinated persons. This form was marked by a fever that occurred after the usual incubation period and could be confirmed only by antibody studies or, rarely, by viral culture.

- Moreover, there were two very rare and fulminating types of smallpox, the malignant (flat) and hemorrhagic forms, which were usually fatal.

<u>Transmission: -</u>

1. Transmission occurred through inhalation of airborne variola virus, usually droplets expressed from the oral, nasal, or pharyngeal mucosa of an infected person.

2. It was transmitted from one person to another primarily through prolonged face-to-face contact with an infected person, usually, within a distance of 1.8 m (6 feet),

3. It could also be spread through direct contact with infected bodily fluids or contaminated objects (fomites) such as bedding or clothing.

4. Rarely, smallpox was spread by virus carried in the air in enclosed settings such as buildings, buses, and trains.

5. The virus can cross the placenta, but the incidence of congenital smallpox was relatively low.
6. Smallpox was not notably infectious in the prodromal period and viral shedding was usually delayed until the appearance of the rash, which was often accompanied by lesions in the mouth and pharynx.

7. The virus can be transmitted most frequently during the first week of the rash when most of the skin lesions were intact.

8. Infectivity waned in 7 to 10 days when scabs formed over the lesions, but the infected person was contagious until the last smallpox scab fell off.

9. Smallpox was highly contagious, but generally spread more slowly and less widely than some other viral diseases, perhaps because transmission required close contact and occurred after the onset of the rash.

10. The overall rate of infection was also affected by the short duration of the infectious stage.

11. In temperate areas, the number of smallpox infections was highest during the winter and spring. In tropical areas, seasonal variation was less evident and the disease was present throughout the year.

12. Age distribution of smallpox infections depended on acquired immunity.

13. Vaccination immunity declined over time and was probably lost within thirty years.

14. Smallpox was not known to be transmitted by insects or animals and there was no asymptomatic carrier state.

VACCINIA VIRUS

- A large, complex, enveloped virus belonging to the poxvirus family.

- It has a linear, double-stranded DNA genome.

-The vaccinia virus is the source of the modern smallpox vaccine, which the World Health Organization used to eradicate smallpox in a global vaccination campaign in 1958–1977.
-Although smallpox no longer exists in the wild, vaccinia virus is still studied widely by scientists as a tool for gene therapy and genetic engineering.
-Vaccinia virus is able to undergo multiplicity reactivation (MR), the process by which two, or more, virus genomes containing otherwise lethal damage interact within an infected cell to form a viable virus genome, MR provides the advantage of recombinational repair of genome damages.

- Whole-genome sequencing has revealed that vaccinia is most closely related to horsepox, and the cowpox strains found in Great Britain are the least closely related to vaccinia.
- Vaccinia contains within its genome genes for several proteins that give the virus resistance to interferons.
- Vaccinia virus infection is typically very mild and often does not cause symptoms in healthy individuals, although it may cause rash and fever.
- Immune responses generated from a vaccinia virus infection protects the person against a lethal smallpox infection. For this reason, vaccinia virus was, and still is, being used as a live-virus vaccine against smallpox.

Questions

1- Compare between HSV type 1 and 2 ?
2- what are the Steps of HSV viral replication?
3- Enumerate the types of human herpes viruses?

LECTURE-7

Hepatitis virus

Hepatitis Viruses:

Hepatitis is a clinical syndrome caused by many pathogens including viruses. There are six medically important viruses that are called hepatitis viruses because their main site of infection is liver. These viruses are hepatitis A virus (HAV), hepatitis B virus (HBV), hepatitis C virus (HCV), hepatitis D virus (HDV), hepatitis E virus (HEV), and newly described G virus (HGV). Although these viruses infect the liver as common target organ, they however, differ greatly in their morphology, replication pattern, and course of infection. Hepatocytes are the primary target of true.

The characteristics of the five known hepatitis viruses are shown in Table 35-1.

TABLE 35-1 Characteristics of Hepatitis Viruses

Virus	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
Family	Picornaviridae	Hepadnaviridae	Flaviviridae	Unclassified	Hepeviridae
Genus	*Hepatovirus*	*Orthohepadnavirus*	*Hepacivirus*	*Deltavirus*	*Hepevirus*
Virion	27 nm, icosahedral	42 nm, spherical	60 nm, spherical	35 nm, spherical	30–32 nm, icosahedral
Envelope	No	Yes (HBsAg)	Yes	Yes (HBsAg)	No
Genome	ssRNA	dsDNA	ssRNA	ssRNA	ssRNA
Genome size (kb)	7.5	3.2	9.4	1.7	7.2
Stability	Heat and acid stable	Acid sensitive	Ether sensitive, acid sensitive	Acid sensitive	Heat stable
Transmission	Fecal–oral	Parenteral	Parenteral	Parenteral	Fecal–oral
Prevalence	High	High	Moderate	Low, regional	Regional
Fulminant disease	Rare	Rare	Rare	Frequent	In pregnancy
Chronic disease	Never	Often	Often	Often	Never
Oncogenic	No	Yes	Yes	?	No

ds, double stranded; HBsAg, hepatitis B surface antigen; ss, single stranded.

These viruses infect the liver and cause distinct clinical pathology by producing characteristic symptoms of jaundice and production and release of liver enzymes in the serum. Most of these diseases spread very fast because infected individuals are contagious not only during stage of manifestation of the disease but also during the phase of incubation. Human infections associated with hepatitis viruses are summarized in Table 66-2.

TABLE 66-2 Diseases associated with hepatitis viruses

Virus	Diseases
HAV	Acute hepatitis
HBV	Acute and chronic hepatitis
HCV	Acute HCV infection, chronic HCV infection, and cirrhosis and other complications
HDV	Acute self-limited infection to acute fulminant liver failure
HEV	Serious infection in pregnant women
HGV	Coinfection in chronic HBV and HCV infection

Most cases of acute viral hepatitis in children and adults are caused by one of the following five agents:

1- Hepatitis A virus (HAV): the etiologic agent of viral hepatitis type A (infectious hepatitis)

2- Hepatitis B virus (HBV): which is associated with viral hepatitis B (serum hepatitis)

3- Hepatitis C virus (HCV): the agent of hepatitis C (common cause of posttransfusion hepatitis)

4- Hepatitis D (HDV): a defective virus dependent on coinfection with HBV

5- Hepatitis E virus (HEV): the agent of enterically transmitted hepatitis Hepatitis viruses produce acute inflammation of the liver, resulting in a clinical illness characterized by fever, gastrointestinal symptoms such as nausea and vomiting, and jaundice.

Hepatitis Type A and E

Viral classification

Hepatitis A Virus

HAV is an **Enterovirus** belonging to the **Picornaviridae** family in the genus **Hepatovirus**. HAV is a non- enveloped virus that has evolved a unique strategy by which it hijacks cellular membranes, exiting the host cells fully cloaked in a lipid membrane.

The viral particle is an **icosahedral** capsid and the genome is a **single-stranded**, **positive-polarity RNA**.

Hepatitis E Virus

HEV belongs to the family **Hepeviridae** in the genus **Hepevirus**.

HEV is **non-** enveloped RNA virus with an **icosahedral** capsid. The genome is a **positive**-sense single-stranded RNA.

Transmission

Both HAV and HEV transmitted to humans fecal-orally, usually through fecally comtaminated food and water.

Clinical Significance

HAV

Acute HAV infection is presented as an acute illness causing jaundice or elevated serum aminotransferases. Incubation period 2-6 weeks. HAV infection may manifest as fatigue, malaise, vomiting, anorexia, fever, and abdominal pain. HAV is not a serious virus and has a vaccine

HEV

The incubation period for hepatitis E ranges from 15 to 40 days. Acute HEV infection is presented as an acute illness causing jaundice or elevated serum aminotransferases. Gastrointestinal symptoms include diarrhea, nausea, hepatomegaly, splenomegaly, and vomiting.

Detection of HAV and HEV

Serologic testing is required for adequate diagnosis for both HAV and HEV. Both could be diagnosed by PCR

Hepatitis B virus

HBV is classified as a genus of **hepadnavirus**, **Hepadnaviridae** family. HBV establishes chronic infections, especially in those infected as infants; it is a major factor in the eventual development of liver disease and hepatocellular carcinoma in those individuals.

Structure and replication of hepatitis B virus

The HBV virion, historically referred to as the "Dane particle," consists of an **icosahedral** nucleocapsid enclosed in an **envelope**.

- **Organization of the hepatitis B virus genome:** HBV DNA genome is a partly doublestranded, circular DNA molecule (that is, one strand is longer than the other) as shown in Figure 26.4. The short "plus" strand is only 50 to 80 percent as long as its complementary strand, the "minus" strand.
- **Viral proteins:** The four proteins encoded by viral DNA are:
  1) the core protein [hepatitis B nucleocapsid core antigen (HBcAg)];
  2) envelope protein [a glycoprotein referred to as hepatitis B surface antigen (HBsAg)];
  3) multifunctional reverse transcriptase/DNA polymerase, which is complexed with the DNA genome within the capsid; and
  4) a nonstructural regulatory protein designated the "X protein B.

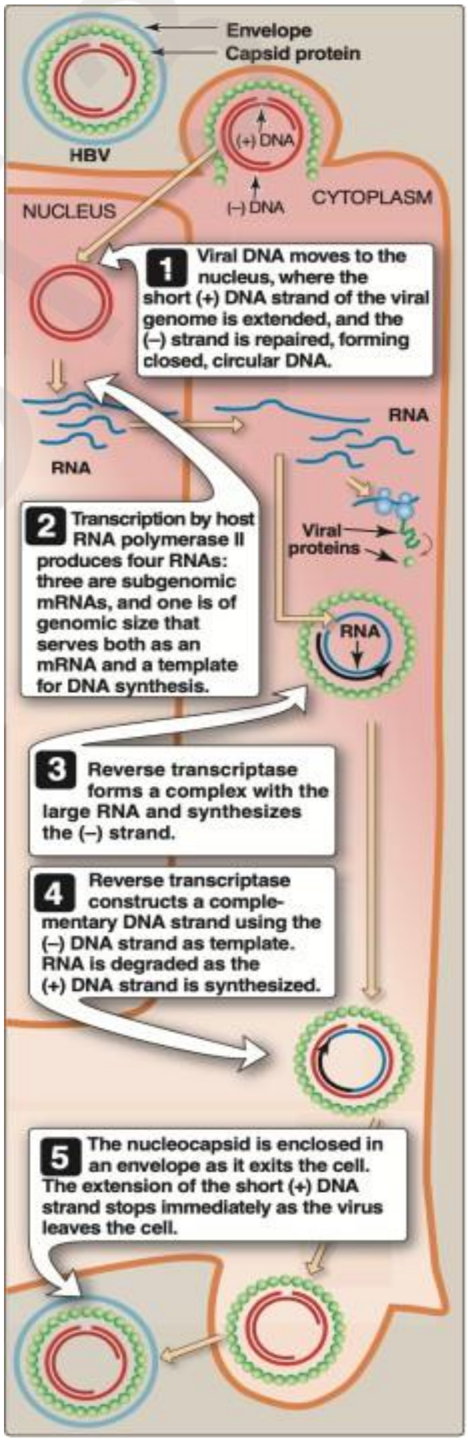


Figure 26.4 Replication of hepatitis B virus (HBV)

Transmission Infectious

HBV is present in all body fluids of an infected individual. Therefore, blood, semen, saliva, and breastmilk, for example, serve as sources of infection. Individuals infected at an early age have an increased risk of developing hepatocellular carcinoma later in life. Hepatitis B is primarily a disease of infants in developing nations, and, in Western countries, it is mostly confined to adults who usually contract HBV infection through sexual intercourse or blood exposure from shared needles during injecting drug use.

Pathogenesis

Hepatocytes are the primary cell type infected by HBV. The primary cause of hepatic cell destruction appears to be the cell-mediated immune response, which results in inflammation and necrosis. The cells involved are cytotoxic T cells, which react specifically with the fragments of nucleocapsid proteins (HBcAg and HBeAg), expressed on the surface of infected hepatocytes.

Clinical significance

Acute disease Chronically infected people serve as the reservoir of transmissible virus in the population. In most individuals, the primary infection is asymptomatic and resolves as a result of an effective cellmediated immune response Phases in acute hepatitis B virus infections: Following infection, HBV has incubation period of between 45 and 120 days. Following this period, a pre-jaundice phase occurs. The acute, icteric (jaundice) phase then follows. During this phase, dark urine, due to bilirubinuria, and jaundice (a yellowish coloration of mucous membranes, conjunctivae, and skin) are evident.

Laboratory identification

The purpose of diagnostic laboratory studies of patients with clinical hepatitis is to, first, determine which hepatitis virus is the cause of the illness and, second (for HBV), to distinguish acute from chronic infections.
Elevations of aminotransferases, bilirubin, and prothrombin time all contribute to the initial evaluation of hepatitis. Commonly known as ELISA, enzyme-linked immunosorbent assay and other immunologic techniques for detection

of viral antigens and antibodies are the primary means to distinguish among HAV, HBV, HCV, and HDV. In addition, identification of the presence or absence of specific antiviral antibodies and viral antigens permits differentiating between acute and chronic HBV infections.

Prevention

The use of a highly effective vaccine has led protection of adults from infection as well as newborns from infection by transmission from HBV- positive mothers.

1. Active immunization: The use of a highly effective vaccine has led protection of adults from infection as well as newborns from infection by transmission from HBV- positive mothers.

2. Passive immunization: Hepatitis B immunoglobulin (HBIG) is prepared from the blood of donors is recommended as the initial step in preventing infection of individuals accidentally exposed to HBV-contaminated blood by needlestick or other means and of those exposed to infection by sexual contact with an HBV-positive partner.

It is also strongly recommended that pregnant women should be screened for HBsAg. Infants born to mothers who are HBV positive are given HBIG plus hepatitis B vaccine at birth, followed by additional doses of vaccine at 1 and 6 months.

Hepatitis D virus (Delta Agent)

HDV is found in nature only as a coinfection with HBV. It is significant because its presence results in more severe acute disease, with a greater risk of fulminant hepatitis and, in chronically infected patients, a greater risk of cirrhosis and liver cancer.

Structure and replication

HDV does not fall into any known group of animal viruses. It has a circular, single-stranded RNA genome with negative polarity. HBV-coded HBsAg. Thus, HDV requires HBV to serve as a helper virus for infectious HDV production. The HDV RNA genome is replicated and transcribed in the nucleus by cellular enzymes.

Transmission and pathogenesis

Because HDV exists only in association with HBV, it can be transmitted by the same routes. However, it does not appear to be transmitted sexually as frequently as HBV or HIV.

Pathologically, liver damage is essentially the same as in other viral hepatitides, but the presence of HDV usually results in more extensive and severe damage.

Laboratory identification

The immunologically based methods used to diagnosis HBV are also applied to HDV. The delta (D) antigen and immunoglobulin M antibodies against it can be detected in serum. The presence of HDV RNA in serum or liver tissue, as detected by hybridization with or without the use of reverse transcriptase and polymerase chain reaction amplification, is an indicator of active infection.

Hepatitis C virus

HCV is a **positive-stranded RNA virus**, classified as family **Flaviviridae**, genus **Hepacivirus**.

Hepatitis C virus genera

Hepatitis C virus (HCV) was discovered in 1988 in the course of searching for the cause of non-A, non-B, transfusion-associated hepatitis. At that time, HCV accounted for 90 percent of the cases of non-A, non-B hepatitis.

Transmission and pathogenesis:

Although HCV was initially identified as a major cause of posttransfusion hepatitis, intravenous drug users and patients on hemodialysis are also at high risk for infection with HCV. Tattooing is also a leading cause of HCV infection. In addition, there is evidence for sexual transmission of HCV as well as for transmission from mother to infant.

In the infected individual, viral replication occurs in the hepatocyte and, probably, also in mononuclear cells (lymphocytes and macrophages). Destruction of liver cells may result both from a direct effect of the activities of viral gene products and from the host immune response, including cytotoxic T cells. HCV have been associated with hepatocellular carcinoma development.

Chronic Hepatitis

The risk of chronic HCV infection is high. - Patients with chronic infection are asymptomatic or have only mild nonspecific symptoms since cirrhosis is not present. The most frequent complaint is fatigue. Less common manifestations are nausea, weakness, myalgia, arthralgia, and weight lose.

Viral coinfection:

HCV progression is more rapid inHIV-infected patients. Acute hepatitis B in a patient with chronic hepatitis C may be more severe. Liver damage is usually worse and progression faster in patients with dual HBV/HCV infections.

Laboratory identification:

A specific diagnosis can be made by demonstration of antibodies that react with a combination of recombinant viral proteins. Sensitive tests are also now available for detection of the viral nucleic acid by RT-PCR.

	Hepatitis A virus (HAV)	Hepatitis B virus (HBV)	Hepatitis C virus (HCV)	Hepatitis D virus (HDV)	Hepatitis E virus (HEV)
Viral family	*picornavirus*	*hepadnavirus*	*flavivirus*	*deltavirus*	*hepevirus*
Chronic	no	yes	yes	yes	no
Vaccine	yes	yes	**NO**	yes	yes

Questions

1. The transmission of HCV will be via

a- Contaminated needle. b- Contaminated food. c- Aerosols d- Contaminated water

2. HBV genome is

a- ssRNA b- dsRNA c- Gaped dsDNA d- ssDNA

3. The transmission of HAV will be via

a- Contaminated needle. b- Contaminated food. c- Aerosols d- Sexual transmission

4. Which virus belongs to Flaviviridae

family? a- HAV b- HBV c- HCV d- HDV

5. Which virus belongs to hepadnaviridae

family? a- HAV b- HBV c- HCV d- HDV

LECTURE- 8

Human Immune Deficiency virus

Human immunodeficiency virus (HIV) Is a retrovirus that causes human AIDS. Many retroviruses infect vertebrates, One genus of retrovirus, Lentivirus, includes the subspecies HIV-1 and HIV-2, which cause AIDS.

Family: Retroviridae

Genus: Lentivirus

Species: Human Immunodeficiency Virus (HIV)

●HIV infects mainly CD4+ T cells, macrophages, and dendritic cells which express the surface receptor CD4.

●Destroying CD4+ T cells leads to opportunistic infection.

● Acquired immunodeficiency syndrome (AIDS) is the end stage of the disease that is associated with CD4+ T cell depletion, multiple or recurrent opportunistic infections, and unusual cancer (Kaposi sarcoma).

●HIV patient is different from an AIDS patient. AIDS is an end stage of HIV virus.

Two types of HIV

A- HIV-1: has 9 subtypes it is responsible for most cases of acquired immunodeficiency syndrome (AIDS)

HIV-1: most common cause of AIDS which :-

- Causes HIV infection worldwide.
- Highly virulent.
- Highly susceptible to mutations.

B- HIV-2: has 5 subtype it is less commonly and less virulent AIDS was first describe as disease in 1981 and the virus was isolated by end of 1983

- HIV-2 Causes the infection in specific regions e.g. West Africa, Mozambique.
- Relatively less virulent so less pathogenic .
- Relatively less susceptible to mutations.

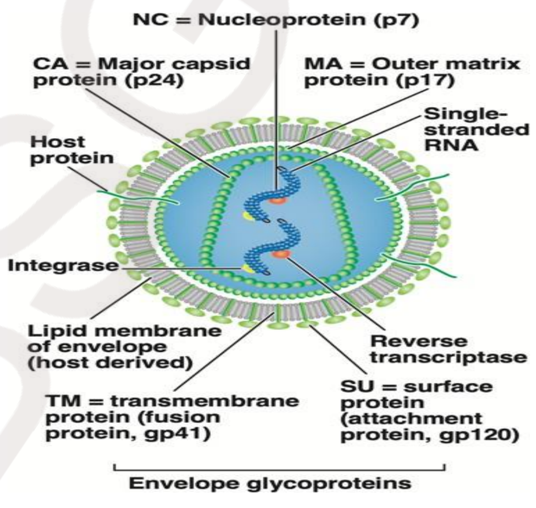


Figure: Structure of the human immunodeficiency virus.

Biological structure of HIV

- Envelope: the virus is surrounding by bilayer lipid envelope that covered by projected spikes (glycoprotein: gp41, gp120) which may act as attachment sites (HIV undergoes from high rate of antigenic variation in envelope glycoproteins). The symmetry of HIV is Icosahedral (20 sided).
- Capsid proteins: consist of several proteins as capsid protein (p24) and matrix protein (p17)
- Core protein: contain many enzymes as reverse transcriptase (R tase) integrase

Enveloped virus and Virion consist of :-

1- Glycoprotein envelope (gp120, gp41).

2- Matrix layer (p17).

3- Capsid (p24).

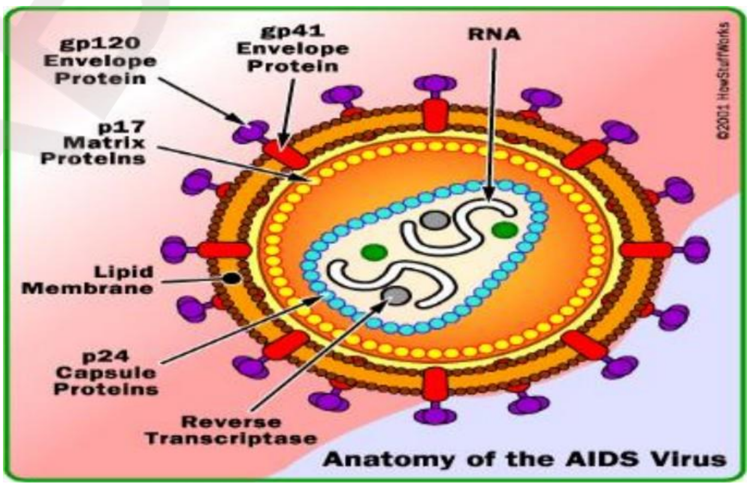
4- Two copies of ss-RNA.

5- Enzymes: • Reverse transcriptase • Integrase, • Protease

6- Genome consists of two copies of (+) ssRNA (diploid).

- **The genome consists of 9 genes:**

3 structural genes (gag, pol, env) required for the replication of retroviruses and 6 non-structural genes (tat, nef, rev, vif, vpr, vpu) regulate viral expression and are important in disease pathogenesis in vivo.



Viral Replication

Viral replication will be described into six steps:

1. Attachment to a specific cell surface receptor

Attachment is accomplished via the gp120 fragment of the env gene product on the HIV surface, which preferentially binds to a CD4 receptor molecule. Thus, the virus infects helper T cells, lymphocytes, monocytes, and dendritic cells, which contain this protein in their cell membranes.

2. Entry of virus into the cell: An additional coreceptor, a chemokine receptor (CCR5 and CXCR4 produced by lymphocytes and macrophages), is required for entry of the viral core into the cell. Binding to a coreceptor activates the viral gp41 gene product, triggering fusion between the viral envelope and the cell membrane.

3. Reverse transcription of viral RNA: After entering the host cell, HIV RNA is transcribed into DNA by reverse transcriptase—an RNAdirected DNA polymerase that enters host cells as part of the viral nucleocapsid.

4. Integration of the provirus into host cell DNA: Viral integrase cleaves the chromosomal DNA and covalently inserts the provirus, the integrated provirus thus becomes a stable part of the cell genome.

5. Transcription and translation of integrated viral DNA sequences: With host cell activation, DNA is transcribed to mRNA and viral genomic RNA. Viral mRNAs are translated to give viral enzymes and structural proteins.

6. Assembly and maturation of infectious progeny: As the virion buds from the surface, the viral protease is activated and cleaves the poly- proteins into their component proteins, which then assemble into the mature virion. The mature retrovirus acquires an envelope as it buds from the infected host cell.

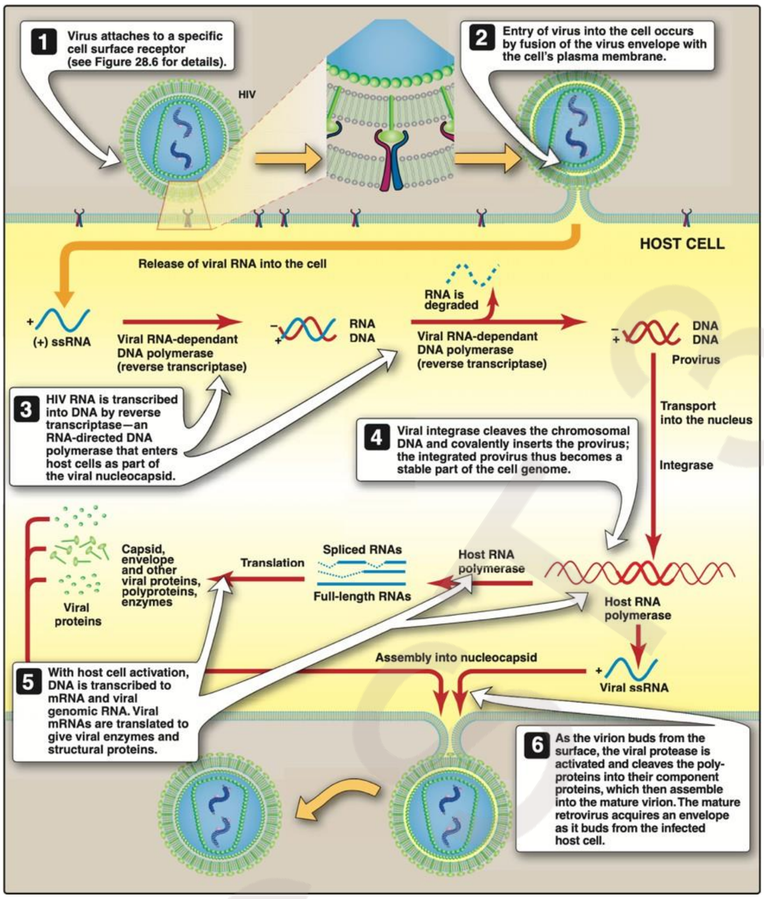


Figure: The human immunodeficiency virus (HIV) replication cycle. ssRNA = single strand RNA.

Virus receptors

The virus use CD4 molecule which is expressed on macrophages and T-lymphocytes in addition to that HIV required a co receptor which is CCR5 a co receptor for the macrophage strains of HIV-1, whereas CXCR4 is the co receptor for the lymphocyte strain of HIV-1.these co receptors are acting normally as a chemokines receptors on the cell, and required for fusion of the virus with the cell membrane . the virus first bind to CD4 and then to the co receptors. These interactions causes conformational changes in the viral envelope activating gp41 fusion protein and triggering membrane cell fusion. Individual who possess homozygous deletions in these co receptors may be protected from infection by HIV-1.

Mod of transmesion

υ **Sexual contact** : HIV has high affinity to semen and vaginal secretion therefore the virus can be transmitted by anal or vaginal intercourse among homosexual and heterosexual individuals

υ **parenteral transmission** : it can be transmitted by blood transfusion or by needles or syringes such as intravascular drugs uses(IVDU) υ mother to child : HIV can transmitted to neonate across placenta or during delivery or breast milk

υ **other methods** : for transmission of HIV fluids or body such as urine tear saliva Bacterial infections such as TB syphilis salmonella infection viral infection EBV CMV hepatitis and herpes simplex fungal infection as candida albicans (cause oral thrush) protozoa infection pneumocystis carinii (cause pneumonia).

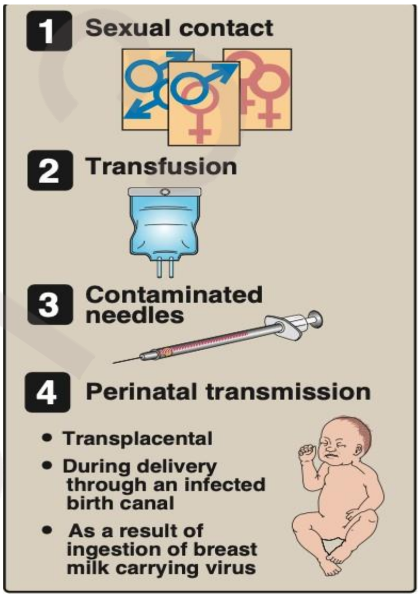


Figure: Common modes of transmission of HIV.

Acute phase of HIV:

●Incubation period 2 weeks and lasts for about 12 weeks.

●Mostly asymptomatic, but in about 25-65% of the cases, patients may develop symptoms resemble infectious mononucleosis or Flu (fever, headache, anorexia, fatigue, lymphadenopathy, skin rash) which resolved in 2 weeks.

● Rapid viral replication (high viral load >106 copies/mL) .

● Gradual decrease in CD4+ T cell count.

● Blood markers in the acute stage:

● Normal to slightly decrease number of CD4+ T cells.

● Appearance of the viral RNA, and then the core antigen (p24 antigen) which indicate active viral replication.

● Appearance of two antibodies, Anti-envelop (Anti-gp120) & Anti-core (Anti-p24).
● The first choice marker for detection HIV in the acute phase is HIV RNA.

● Antibody tests may give false negative (no antibodies were detected despite the presence of HIV) results during the window period, an interval of three weeks to six months between the time of HIV infection and the production of measurable antibodies to HIV seroconversion.

Chronic phase of HIV:

Lasts for about 10 years in adults, and 5 years in children.

●Totally asymptomatic but the patients is still contagious.

●Relatively low viral load (500 cells/mm3).

●At the end of this stage, two syndromes appear:

1- Persistent generalized lymphadenopathy (PGL):

→Is defined as enlargement of lymph nodes for at least 1 cm in diameter in the absence of any illnesses or medications that known to cause PGL. → Clinical features: • In two or more lymph nodes out of the inguinal area. → Persists for at least 3 months.

2- AIDS-related complex (ARC):

►Is a group of clinical symptoms that come before AIDS and may include the following:

► Fever of unknown origin that persists > 1 month.

► Chronic diarrhea, persisting > 1 month.

►Weight loss > 10% of the original weight (slim disease).

►Fatigue, night sweating, and malaise.

►Neurological disease as myelopathy and peripheral neuropathy.

Blood markers in the chronic stage:

- Viral load (HIV RNA) increases gradually, and HIV core antigen (p24) may appear in blood.
- Anti-envelop (Anti-gp120) & Anti-core (Anti-p24) are positive.
- CD4+ T cell count gradually decreased .

AIDS phase

The end stage of the disease.

-Continuous viral replication (high viral load).

-Marked decrease in CD4+ T cell count < 200 cell/mm3.

-Defects in cellular immunity. Persistent or frequent multiple opportunistic infections.

-Unusual cancer (i.e. Kaposi sarcoma).

Blood markers in AIDS stage:

- High viral load (HIV RNA), and HIV core antigen (p24) appears in blood.
- Detection of both HIV RNA & the antigen p24 indicative of active viral replication.
- Anti-envelop (Anti-gp120) & Anti-core (Anti-p24) are positive.
- CD4+ T cell count decreased to very low levels (<200 cells/mm3).

Dignoses of HIV

1. Cell count of WBC for determination of T4/T8 ratio υ Isolation of virus by cell culture (in difficult)
2. serologic for detection of HIV an tibodies by ELISA radioimmune assay (RIA) and immunofluorescent test (IFT)
3. PCR has highly sensitivity and specificity to detect HIV genome in infected Cell.

Control

1.prevention : safety sex practice

2. routine screening of donated blood for HIV

3. needles exchange program for IVDUS teeth brush razor …..etc should be not Used

4. avoidance pregnancy breast –feeding if infected mother

Treatment with antiviral therapy

υ Viral binding inhibitors such as CD4-lgG chimera

υ Rtase inhibitoors :azidothymidine (AZT) υ protein synthesis inhibitors ritonavir

υ Viral assembly inhibitors interferon –Alfa υ Gene therapy is developed now. υ No vaccine available (because changes in antigenicity of HIV)

Questions

Q1: Define AIDS ?

Q2:- Answer True or false :-

1- HIV patient is different from an AIDS patient. AIDS is an end stage of HIV virus.

2- HIV belong to the lentiviridae .

Q3:- Multiple choice :-

1- Co receptor for the macrophage strains of HIV-1 is :-

a- CD4 b- CCR5 c- CD8 d- CXCR4

2- The 1st choice marker for detection HIV in the acute phase is :-

a- p24 antigen b- Anti p24 c- HIV RNA d- Anti-gp120

3- AIDS stage in HIV infection include :-

a- Normal CD4+ T cell count b- high CD4+ T cell count c- decrease CD4+ T cell count d- None of them.

LECTURE-9

Orthomyxoviruses

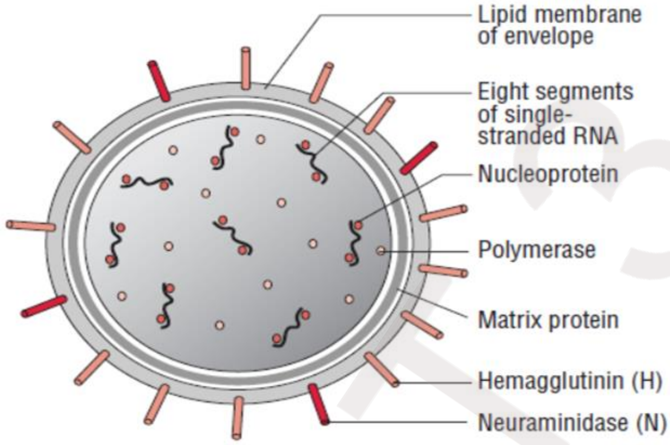
The term myxovirus was coined for a group of enveloped RNA viruses that have the ability to adsorb onto mucoprotein receptors on erythrocytes, causing hemagglutination.
It included influenza, mumps, parainfluenza, and Newcastle disease viruses.

Properties	Orthomyxoviruses	Paramyxoviruses
Size	80–120 nm	100–300 nm
Shape	Spherical	Pleomorphic
Genome	Segmented—eight pieces of RNA	Single, linear RNA
Gene reassortment	Common	Not reported
Antigenic stability	Variable	Stable
Hemolysis	Absent	Present
Site of synthesis of ribonucleoprotein	Nucleus	Cytoplasm
DNA-dependent RNA synthesis	Required for multiplication of the virus	Not required for the multiplication of the virus

Differences between orthomyxoviruses and paramyxoviruses

Influenza Viruses:
Influenza viruses are classic respiratory viruses. They cause influenza, an acute respiratory disease, with well-defined systemic symptoms. Influenza is an acute infectious disease of the respiratory tract that occurs in sporadic, epidemic, and pandemic forms.
Properties of the Virus:
Influenza viruses show following features:
■ Influenza viruses are spherical or filamentous, enveloped particles 80–120 nm in diameter.
■ Influenza virus is composed of a characteristic segmented single-stranded RNA genome, a nucleocapsid, and anenvelope.
■ The viral genome is a single-stranded antisense RNA. The genome consists of an RNA-dependent RNA polymerase, which transcribes the negative-polarity genome into mRNA. The genome, therefore, is not infectious. The viral RNA has a molecular weight of 5 million daltons and a length of 13,600 nucleotides. Characteristically, it is segmented and consists of seven or eight segments. These segments code for different proteins which are NS1, NS2, NP, M1, M2, M3, HA, and NA.
■ The genome is present in a helically symmetric nucleocapsid surrounded by a lipid envelope. The envelope has an inner membrane protein layer and an outer lipid layer. The membrane proteins are known as matrix or M protein and are composed of two

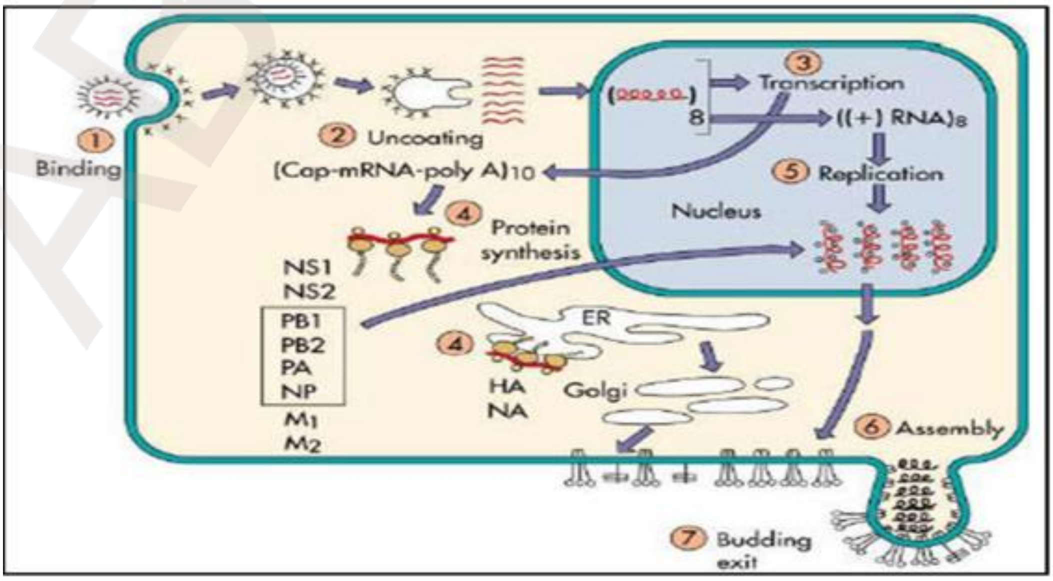
components M1 and M2.
■ Two types of spikes or peplomers project from the envelope:
(a) the triangular hemagglutinin (HA) peplomers and (b) the mushroom-shaped neuraminidase (NA) peplomers.



Structure of an influenza virus

Influenza virus life cycle:

Replication of influenza A virus include after binding to sialic acid-containing receptors, influenza is endocytosed which fuses with the vesicle membrane and uncoats mediated by the M2 proteins and is facilitated by the low pH within the endosome/vesicle. The viral nucleocapsid enters the cytoplasm and migrates to the nucleus where the genome RNA (8 segments) gets transcribed into mRNA by the viral RNA polymerase (transcriptase). Unlike for most other RNA viruses, transcription and replication of the genome occur in the nucleus were Viral proteins synthesized } Most RNA's move to cytoplasm, some remain in the nucleus to serve as a template for the synthesis of negative polarity strand RNA genomes for the progeny, by a different subunit of viral RNA polymerase (replicase). Helical nucleocapsid segments form and associated with the M1 protein-lined membranes containing M2 and the HA and NA glycoprotein's. The virus buds from the plasma membrane.



Influenza virus life cycle

Antigenic Changes of Orthomyxoviruses:

Changes in the antigenicity of hemagglutinin and neuraminidase confers on the Influenza A virus the ability to cause pandemics.

Two types of antigenic changes are known

1- Antigenic drift refer to a minor change based on accumulate mutations during virus replication in the genome RNA. Thus, Influenza viruses have many serotypes.

2- Antigenic shift that involves a major change based on the reassortment of segments of the genome RNA.

Antigenic shifts can result from mechanisms Genetic reassortment between subtypes. Reassortment is possible whenever two different influenza viruses infect a cell simultaneously; when the new viruses (the progeny) are assembled, they may contain some genes from one parent virus and some genes from the other.

Types of influenza viruses

There are four types of influenza viruses: A, B, C and D

1. Influenza A viruses

Influenza A viruses include the avian, swine, equine and canine influenza viruses, as well as the human influenza A viruses. Influenza A viruses are classified into subtypes based on two surface antigens, the hemagglutinin (H) and neuraminidase (N) protein. There are 18 different known H antigens (H1 to H18) and 11 different known N antigens (N1 to N11). H1N1, H1N2, and H3N2 are the only known influenza A virus subtypes currently circulating among humans.

2. Influenza B viruses Influenza B viruses are mainly found in humans. These viruses can cause epidemics in human population, but have not, to date, been responsible for pandemics.

3. Influenza C viruses Influenza type C infections generally cause mild illness and are not thought to cause human flu epidemics. 4- Influenza D viruses primarily affect cattle and are not known to infect or cause illness in people. Viral Transmission Influenza viruses are transmitted in aerosols created by coughing and sneezing, and by contact with nasal discharges, either directly or on fomites. Close contact and closed environments favor transmission. Person-to-person transmission occurs with the H1N1 virus that is currently circulating in humans.

Clinical findings:

Incubation Period : The incubation period for human influenza is usually short; most infections appear after one to four days. The incubation period for the novel H1N1 virus circulating in humans appears to be 2 to 7 days. Clinical Signs &Pathogenicity Uncomplicated infections with human influenza A or B viruses are usually characterized by upper respiratory symptoms, which may include fever, chills, anorexia, headache, myalgia, weakness, sneezing, rhinitis, sore throat and a nonproductive cough. Nausea, vomiting and otitis media are common in children, and febrile seizures have been reported in severe cases. Most people recover in one to seven days, but in some cases, the symptoms may last up to two weeks or longer. More severe symptoms, including pneumonia, can be seen in individuals with chronic respiratory or heart disease. Secondary bacterial or viral infections may also occur.

Laboratory Diagnosis of Human Influenza:

Specimen collection Respiratory specimens: Respiratory specimens obtained within four days of onset of symptoms and different types of respiratory specimens can be used such as nasal washes and nasopharyngeal aspirates tend to be more sensitive than pharyngeal swabs.

Blood specimens: Acute and convalescent serum samples 14 − 21 days should be collected to demonstrate a significant (at least fourfold) rise in strain-specific antibody titer. Laboratory Tests

1- Isolation methods (Viral Culture) - Embryonated egg culture - Cell culture: - Various cell-lines are utilized to isolate influenza viruses, most commonly primary monkey kidney cells. Infection of cells gives a visible cytopathic effect (CPE).

2- Direct methods

- Immunofluorescence
- Enzyme immuno assays
- Reverse transcription polymerase chain reaction (RT-PCR).

3- Serology Different serological techniques are available for influenza diagnosis includes haemagglutination inhibition (HI), complement fixation (CF), enzyme immunoassays (EIA) and indirect immunofluorescence.

Post test

Q: - Multiple choice :-

1- Replication of the genome in influenza viruses take place in: -
a-Cytoplasm
b- nucleus
c- cytoplasm and nucleus
d- Golgi apparatus

2- One of influenza virus protein which has ability to inhibit interferon mRNA is:-
a- M1
b-M2
c- HA
d- NS1

3-Antigenic shift in influenza virus refer to: -
a- Minor change in genome
b- Major change in genome
c- Intermediate change in genome
d- None of them

4- All of the following include Neuraminidase (NA) functions except one: -
a- It is a sialidase enzyme that removes sialic acid from glycoconjugates.
b- helps prevent self-aggregation of virions.
c- It facilitates release of virus particles from infected cell surfaces during the budding process
d- Agglutinates red blood cells, and this is the basis of diagnostic Hemagglutination inhibition test.

LECTURE 10

Paramyxoviruses

Paramyxoviruses are the major respiratory pathogens in this age group. The paramyxoviruses include the most important agents of respiratory infections of infants and young children (respiratory syncytial virus [RSV] and the parainfluenza viruses) as well as the causative agents of two of the most common contagious diseases of childhood (mumps and measles). All members of the Paramyxoviridae family initiate infection via the respiratory tract. Whereas replication of the respiratory pathogens is limited to the respiratory epithelia, measles and mumps become disseminated throughout the body and produce generalized disease.

Classification

The family Paramyxoviridae consists of three important genera 1- Paramyxovirus includes parainfluenza and mumps virus. 2- Morbillivirus includes the measles virus. 3-Pneumovirus includes respiratory syncytial virus (RSV), which is responsible for majority of acute respiratory infections in infants and children.

Paramyxovirus Family

GENUS	MEMBERS	GLYCOPROTEINS
Paramyxovirus	mumps human parainfluenza viruses (HPIV 1-4)	HN, F
Morbillivirus	Measles	H, F
Pneumovirus	Respiratory syncytial virus	G, F

Properties of Paramyxo viruses

Virion: Spherical, pleomorphic,150 nm or more in diameter (helicalnucleocapsid,13–18nm)

● Composition: RNA(1%),protein (73%),lipid (20%), carbohydrate(6%)

● Genome: Single- stranded negative RNA, linear, non-segmented, about15kb, no reassortment .

●Proteins: Six to eight structural proteins.

●Envelope: Contains viral glycoprotein (G ,H, or HN)(which sometimes carries hemagglutinin or neuraminidase activity) and fusion(F) glycoprotein

● Replication: Cytoplasm: particles bud from plasma membrane. . A large excess of nucleocapsids are produced in infected cells, which form characteristic cytoplasmic inclusion bodies.

● Outstanding characteristics: Antigenically stable Particles are labile yet highly infectious.

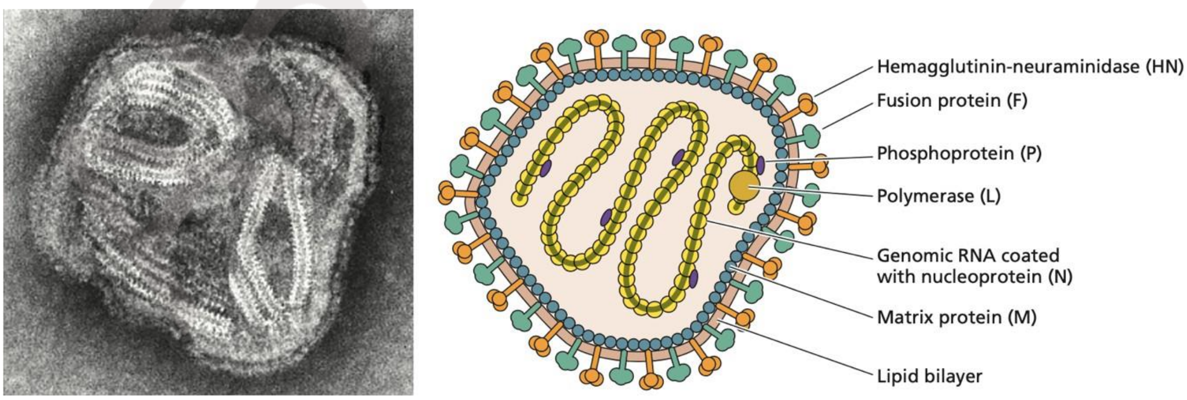


Figure: Paramyxovirus structure

Transmission :-

spread by droplets from the nose and mouth to close contacts. Many of them are highly infectious and go around the community in epidemics- often seasonal, eg. Winter coughs and colds. Fomites might also assist spread

Human parainfluenza viruses (HPIVs)

HPIVs are single-stranded, enveloped RNA viruses of the Paramyoviridaie family. There are four serotypes(1-4) which cause respiratory illnesses in children and adults. HPIVs bind and replicate in the ciliated epithelial cells of the upper and lower respiratory tract and the extent of the infection correlates with the location involved. Seasonal HPIV epidemics result in a significant burden of disease in children and account for 40% of pediatric hospitalizations for lower respiratory tract illnesses and 75% of croup cases.

Pathogenesis of human parainfluenza virus (HPIV) infection

The virus adsorbs to the respiratory epithelial cells by specifically combining with neuraminic acid receptors in the cell through its hemagglutinin. Subsequently, the virus enters the cells following fusion with the cell membrane, mediated by F1 and F2 receptors. The virus replicates rapidly in the cell cytoplasm and causes formation of multinucleated giant cells. The virus also causes the formation of single and multilocular cytoplasmic vacuoles and basophilic or eosinophilic inclusions. The virus causes inflammation of the respiratory tract, leading to secretions of high level of inflammatory cytokines, usually 7–10 days after initial exposure. Airways inflammation, necrosis, and sloughing of respiratory epithelium, edema, and excessive mucus production are the noted pathological features associated with HPIV infections.

Clinical feature

Human parainfluenza viruses cause croup (a heterogeneous group of illnesses that affects the larynx, trachea, and bronchi. The condition manifests as fever, cough, laryngeal obstruction), pneumonia, bronchiolitis and tracheobronchitis, and Otitis media, pharyngitis, conjunctivitis. The severity of the disease occurred in infant **less than 6 months**.

Laboratory Diagnosis

Respiratory specimens include nasopharyngeal aspirations nasal washings, and nasal aspirations.

1- Direct antigen detection The ELISA, immunofluorescence assay are used to detect HPIV antigen

2-Molecular Diagnosis : polymerase chain reaction (PCR) has been developed for detection of HPIV-1, HPIV-2, and HPIV-3 genome in clinical specimens.

3- Isolation and identification Nasal wash are good specimens, culture in monkey kidney cell line , the diagnosis depending on hem adsorption

Prevention and Control

Currently there is no vaccine against infection by HPIV, However, researchers are trying to develop one.

Mumps

Mumps is an acute contagious disease characterized by nonsuppurative enlargement of one or both salivary glands. Mumps virus mostly causes a mild childhood disease, but in adults complications including meningitis and orchitis are fairly common. More than one-third of all mumps infections are asymptomatic.

Figure: Child with mumps (swollen parotid gland)

Pathogenesis & Pathology

Humans are the only natural hosts for mumps virus. Primary replication occurs in nasal or upper respiratory tract epithelial cells. Viremia then disseminates the virus to the salivary glands and other major organ systems. Involvement of the parotid gland is not an obligatory step in the infectious process. The incubation period may range from 2 to 4 weeks but is typically about 14– 18 days. Virus is shed in the saliva from about 3 days before to 9 days after the onset of salivary gland swelling. About one-third of infected individuals do not exhibit obvious symptoms (in apparent infections) but are equally capable of transmitting infection. Virus frequently infects the kidneys and can be detected in the urine of most patients. Viruria may persist for up to 14 days after the onset of clinical symptoms. The central nervous system is also commonly infected and may be involved in the absence of parotitis.

Clinical Findings

Fever, malaise followed by rapid enlargement of the parotid gland and it is painful . mumps may be associated with aseptic meningitis . testis and ovaries may be infected especially after puberty and it may pass to sterility in man but it is rare (not more than 1%).

Laboratory diagnosis of Mumps virus

1) Clinical feature

2) Isolation and identification

The most appropriate clinical samples for viral isolation are saliva, cerebrospinal fluid, and urine collected within a few days after onset of illness. Virus can be recovered from Culture in monkey kidney cells and diagnosis by using the urine for up to 2 weeks. mumps specific antiserum by immunofluorescence method,hemadsorption test can also be used.

3) Nucleic acid detection:- by PCR test.

4) Serology IgM and IgG Abs detection by ELISA and Heamagglution inhibition test.

Treatment and Prevention

● There is no specific therapy .

● Immunization with attenuated live mumps virus vaccine is the best approach to reducing mumps-associated morbidity and mortality rates. Mumps vaccine is available in combination with measles and rubella (MMR) live-virus vaccines.

Measles

Measles is an acute, highly infectious disease characterized by fever, respiratory symptoms, and a maculopapular rash. Complications are common and may be quite serious.

Pathogenesis & Pathology

Humans are the only natural hosts for measles virus. The virus gains access to the human body via the respiratory tract, where it multiplies locally; the infection then spreads to the regional lymphoid tissue, where further multiplication occurs. Primary viremia disseminates the virus. Finally, a secondary viremia seeds the epithelial surfaces of the body, including the skin, respiratory tract, and conjunctiva, where focal replication occurs. The described events occur during the incubation period, which typically lasts 8–12 days but may last up to 3 weeks in adults. involvement of the central nervous system is common in measles. Clinical Findings Infections in non immune hosts are almost always symptomatic. After an incubation period of 8–12 days, measles is typically a 7-11days illness. The prodromal phase is characterized by fever, sneezing, coughing, running nose, redness of the eyes, Koplik spots, and lymphopenia. The conjunctivitis is commonly associated with photophobia. Koplik spots are small, bluish-white ulcerations on the buccal mucosa opposite the lower molars. These spots contain giant cells and viral antigens and appear about 2 days before the maculopapular rash. . The most common complication of measles is otitis media (5–9% of cases). Pneumonia is the most common life-threatening complication of measles, caused by secondary bacterial infections.

Subacute sclerosing panencephalitis (SSPE) is a very rare, but fatal disease of the central nervous system that results from a measles virus infection acquired earlier in life SSPE generally develops 7 to 10 years after a person has measles.

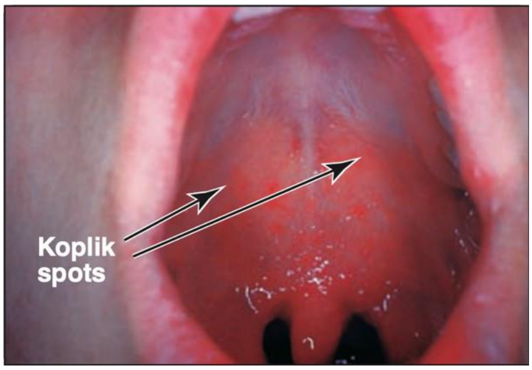


Figure: Koplik spots in the mouth caused by measles virus

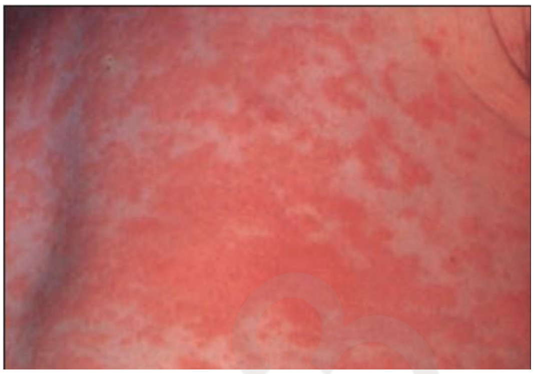


Figure: The measles skin rash (maculopapules)

<u>Laboratory Diagnosis</u>

1) Clinical feature

2) Isolation & Identification of Virus

● Nasopharyngeal and conjunctival swabs, blood samples, respiratory secretions, and urine collected from a patient during the febrile period are appropriate sources for viral isolation. culture in monkey and human kidney cells , diagnosis by cytopathic effect , multinucleated and intra nuclear and intra cytoplasmic inclusion bodies.

3- Antigen detection Measles antigen can be directly detected from specimen include respiratory secretion , nasopharynx and conjunctiva by Immunofluorescence test.

4- Serology IgM and IgG antibodies by ELISA and Heamagglution inhibition test (HI) test.

5- Detection of viral RNA by RT-PCR Is a sensitive method that can be applied to a variety of clinical samples for measles diagnosis. Treatment, Prevention, & Control No treatment . A highly effective and safe attenuated live measles virus vaccine has been available since 1963.

Respiratory syncytial virus(RSV)

It is the most common cause of lower respiratory tract illness in infant and young children.

Pathogenesis and pathology

Replication of the virus occurred initially in the nasopharynx , then the virus may spread to the lower respiratory tract and produce bronchiolitis and pneumonia. The incubation period 3-5 days and virus shedding for 1-3 weeks.

Clinical findings

Common cold, pneumonia in infant and may bronchitis and bronchiolitis which Lifethreatening disease in infant especially under 6, and can lead to chronic lung disease in later life. Reinfection is common in both children and adult with less severity. This virus are a common cause of otitis media about 30% of otitis media cause in infant .

Laboratory diagnosis of Respiratory syncytial virus(RSV)

1- Clinical feature

2- Antigen detection Nasal wash or aspirate are good sample . Virus antigens detection by immunofluorescence test .

3- Isolation and identification of the virus By culturing the specimen into human heteroploid cell line (Hela) and Hep-2, the diagnosis is depend on the cytopathic effect and appearance of giant cells.

4- Nucleic acid detection Diagnosis by detection of the RNA of the virus by PCR.

5- Serology Detection of serum antibodies which include IgM and IgG Abs by using immunofluorescence test . Treatment Supportive care , Ribavirin may be used in the treatment of sever cases by aerosol for 3-6 dayes . No vaccine is available toda but passive immunization immunoglobulin can be given for infected premature infants.

Questions

Q1:- Multiple choice :-

1- Croup caused by:-

a- Measles virus b- Mumps virus c- RSV d- Parainfluenza virus

2- Koplik spots occur in :-

a- Mumps b- RSV c- Measles d- Parainfluenza virus

3- SSPE is complication of :- a- Parainfluenza virus b- Measles c- Mumps d- RSV

4- Orchitis caused by :-

a- Measles virus b- RSV c- Mumps virus d- Parainfluenza virus

5-Which the following of paramyxoviruses belong to pneumovirus ?

a-measles virus, b-parainfluenza , c- mumps virus , d- respiratory syncytial virus (RSV)

LECTURE 11.

Enteric viruses (Polio Virus, Rota virus and Rio)

Characteristic properties:

- Small, non-enveloped viruses

-Multiply in the gut mucosa

-Transmitted from person to person by the fecal-oral route (ingestion disease),

- Spread throughout the body via the blood stream

-Most infections occur during childhood, and they are usually transient but produce lifelong immunity,

-Clinical syndromes are generally mild

-Infections may cause serious disease e.g. paralytic poliomyelitis, meningitis, or myocarditis.

- There is a high degree of serological cross reactivity between the 72 members.

CLASSIFICATION:

Viruses belong to the family Picornaviridae (pico=small - RNA viruses)

1. Enteroviruses:

- Polio 1, 2, 3
- Coxsackie A 1-24
- Coxsackie B 1-6
- ECHO 1-34
- Entero 68-71
- Entero 72 (Hepatitis A)

2. Rhino viruses: > 120 serotypes

3. Other animal viruses: e.g., Foot & Mouth Disease virus.

POLIOVIRUS AND POLIOMYELITIS

General Characters: -

Small (30nm) and stable; an icosahedral capsid enclosing a positive-sense, single-stranded RNA genome. Relatively resistant to extremes of pH and temperature, and to lipid solvents and detergents.

Types: 3 types can be distinguished by antigenic properties. (There are three serotypes poliovirus type-l is most common and it causes most epidemics, poliovirus type-2 is usually associated with epidemics. Poliovirus type-3 occasionally causes epidemics)

CLINICAL FEATURES: -

Source: Only known source is infected man.

Incubation: After ingestion of the virus, there is local multiplication in the oropharynx and associated lymph nodes, the gut mucosa and regional lymph nodes. Thereafter a viraemia follows, and the patient may experience a fever about a week after exposure.

Illness: -

1. Most infections are asymptomatic, although in some there is a minor transient febrile illness.

2. Occasionally (between 1/100 and 1/1000 of cases) the viraemia may lead to CNS involvement and paralysis due to permanent damage to the spinal cord.

3. The patient may experience degrees of headache, fever, meningism, aseptic meningitis, muscle pains, and finally muscle paralysis, usually asymmetrical.

4. Paralysis develops more frequently in adults.

5. The spinal cord may be damaged in a progressive manner from distal to more central with consequent respiratory paralysis and death, or life on a respirator.

6. Virus is produced and released into the gut (and throat initially) and can be isolated from the throat or stools for some weeks following the incubation period.

7. No true long term carrier status occurs.

8. The host's antibody response begins soon after the viraemia.

9. Good solid lifelong immunity results to the specific strain of poliovirus, but subsequent infection with other strains may still occur.

Laboratory Diagnosis: -

(1) Demonstration of the virus: -

a. Virus may be recovered from faeces (also throat swabs), by inoculation of cell cultures and recognition of cytopathic effects with confirmation by neutralization of infectivity with specific antisera.

b. Vaccine strains may be recovered and need to be differentiated from wild strains by molecular nucleic acid techniques (PCR).

Multiple specimens over several days improves chances of recovery of the virus.

(2) Serology (Host immune response):

a. Most cases of poliomyelitis present with paralysis, i.e., quite late in the pathogenesis, and antibodies have already been formed.

b. antibodies are not usually helpful in providing a positive diagnosis of poliomyelitis.

c. detection of specific IgM has not been applied to polio diagnosis.

d. antibodies are traditionally tested by micro-neutralization of infectivity in vitro using antisera to known virus strains

CSF: Polio virus is never found in the CSF but antibodies here mean either CNS infection or a leak from blood antibodies.

Polio Vaccines: -

(1) Live attenuated virus (SABIN) (1963).

2) Killed whole virus (SALK) (1957).

Polio is controlled (eliminated) by: -

(1) Education

(2) Vaccination

(3) Surveillance

ENTEROVIRUSES - OTHER THAN POLIOVIRUS

Coxsackie, Echo, Entero 68-72: -

Virus structure, Epidemiology, Pathogenesis of all the enteroviruses is remarkably similar.

Most infections are silent. Viraemia may lead to involvement of secondary 'target organs' and clinical symptoms and signs related to those organs. Viral meningitis resolves spontaneously without treatment but bacterial meningitis is a medical emergency requiring treatment.

Enteroviruses may be found in the gut of healthy as well as sick children; the association with any illness may be purely co-incidental.

VIRAL GASTEROENTERITIS

Pediatric diarrhea remains one of the major causes of death in young children. The main factors for high incidence and mortality are unsafe water or inadequate sanitation.

The immediate causes are often of an infectious nature (bacteria, parasites, viruses).

A number of different viruses cause diarrhea, of which the most important is the family of Rotaviruses.

1. Rotaviruses have been estimated to cause 30-50% of all cases of severe diarrheal disease in man.

2. Some strains of adenovirus have also been associated with diarrheal disease.

3. A group of "small round viruses have been linked by genetic techniques as closely related to the "Norwalk" agent.

Astroviruses, Coronaviruses, Toroviruses are also associated with gastroenteritis in humans.

REOVIRUSES

General Features and Disease: -

-A group of viruses which have a wide host range, including vertebrates, invertebrates, plants, protists and fungi.

-They lack lipid envelopes and package their segmented genome within multi-layered capsids.

-Reoviruses can affect the gastrointestinal system (such as rotaviruses) and respiratory tract

-The name "reo-" is an acronym for "respiratory enteric orphan" viruses.

-The term "orphan virus" refers to the fact that some of these viruses have been observed not associated with any known disease. Even though viruses in the family Reoviridae have more recently been identified with various diseases, the original name is still used.

-Reovirus infections occur often in humans, but most cases are mild or subclinical. Rotaviruses, however, can cause severe diarrhea and intestinal distress in children, and laboratory studies in mice have implicated orthoreoviruses in the expression of coeliac disease in pre-disposed individuals.

The virus can be readily detected in feces, and may also be recovered from pharyngeal or nasal secretions, urine, cerebrospinal fluid, and blood.

-Despite the ease of finding reoviruses in clinical specimens, their role in human disease or treatment is still uncertain.

-Some viruses of this family, such as phytoreoviruses and oryzaviruses, infect plants. Most of the plant-infecting reoviruses are transmitted between plants by insect vectors. The viruses replicate in both the plant and the insect, generally causing disease in the plant, but little or no harm to the infected insect.

ROTAVIRUSES - (REO virus family)

General Characters: -

Particles are 70 nm round, non-enveloped, double shelled, enclosing a genome of 11 segments of double stranded RNA.

The virus is hardy and may even survive in sewage, despite stringent treatment.

Human rotavirus has proved difficult to culture in vitro, but the serologically related monkey and calf rotaviruses grow easily in cell culture.

Clinical Features: -

Essentially an ingestion disease (faecal-oral route).

Incubation is short: 1 to 3 days.

Illness: Sudden onset watery diarrhea, with or without vomiting. May last up to 6 days (or longer if immunocompromised). The disease is self-limiting.

Complications: Dehydration may result, this can be severe and life threatening in young children.

Laboratory Diagnosis: -

Detection of virus in stools (peaks at day 3 or 4 of diarrhea):-

1. Latex agglutination.

2. Elisa.

3. Electron Microscopy (labour intensive, relatively insensitive).

4. Electrophoresis of RNA segments.

5. Antibody can be detected but is not clinically useful.

Prevention: -

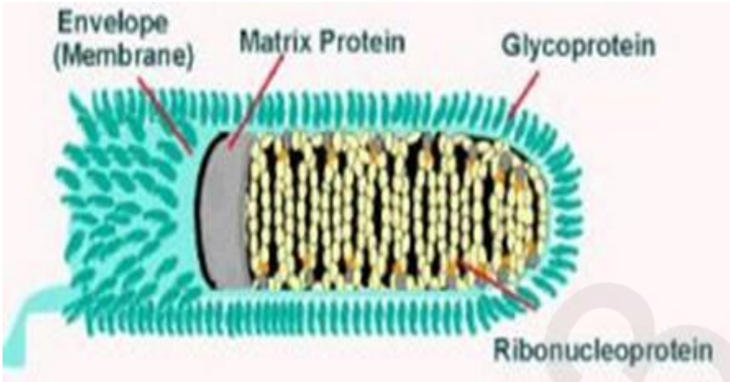
Improved hygiene, education, clean water, specific - breast feeding helps to provide passive immunity in the newborn (from maternal antibodies), oral rehydration, development of a vaccine for Rota virus infection. The prevention of severe dehydration is the main aim, rather than totally preventing infection.

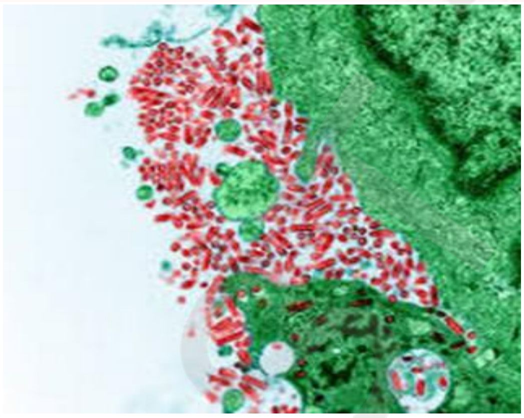
LECTURE-12

-Rabies virus

Classification and General Characters: -

- Member of the Lyssavirus of the Rhabdoviridae.
- ssRNA enveloped virus, helical symmetry.
- Infectivity destroyed by lipid solvents.
- 6-7 nm spike projections are present on the envelope.
- Characteristic bullet-shaped appearance.
- virion 130-240nm.
- -ve stranded RNA codes for 5 proteins.
- Members of the family Rhabdoviridae exceeding wide range of hosts including vertebrates, invertebrates, and plants.
- Rabies virus has been adapted to growth in a wide variety of primary and continuous cell systems.
- The virus is grown in human diploid cells for the purpose of producing a vaccine. It has also been adapted to growth in avian embryos





<u>Pathogenesis and Clinical Features: -</u>

Human infection is usually caused by the bite of dogs or other animals, the virus is present in the saliva of the animals, the disease can also be caused by licks or aerosols.

Once bites take place the virus (in animal saliva) enter deep into the muscle and start multiplying in both muscle tissue and connective tissue then reaching nerves, nerve cells, and finally the brain producing Negri bodies as round or oval inclusions within the cytoplasm of nerve cells of infected human. Negri bodies act to concentrate viral proteins, cellular factors and nucleic acids to build a platform facilitating viral replication. They might also prevent the activation of host innate immunity and restrain the access of viral machineries to cellular antiviral proteins

Five general stages of rabies are recognized in humans:

1-incubation period: - usually 30 to 90 days but ranging from as few as 5 days to longer than 2 years after initial exposure.

2-Prodromal period, which usually lasts from 2 to 10 days, the symptoms are often nonspecific including fever, nausea, vomiting, headache, fatigue, sore throat, cough.

3-Acute neurological period (2 to 3 days, rarely up to 6 days).

4-coma.

5- death.

The incubation period is highly variable, ranging from 7 days to several years. It depends on several factors such as: -

1. Dose of inoculum

2. The severity of the wound

3. The length of the neural path from the wound to the brain, e.g., wounds on the face have a shorter incubation period than wounds in the leg.

Laboratory Diagnosis: -

The diagnosis of animal and human rabies can be made by several methods: -

(1) histopathology (detection of Negri bodies).

(2) virus cultivation.

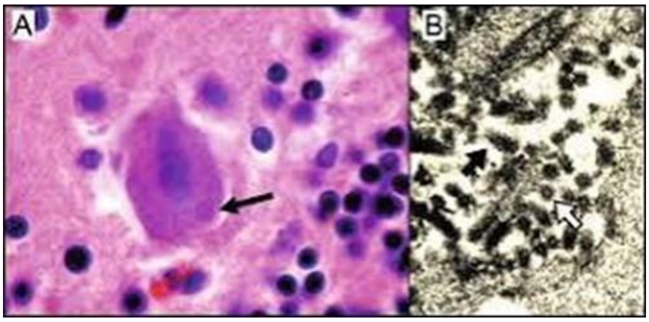
(3) Serology (The most commonly used serological tests were the mouse infection neutralization test (MNT) or the rapid fluorescent focus inhibition test (RFFIT). These tests have now been largely replaced by EIAs. Serology had been reported to be the most useful method for the diagnosis of rabies.

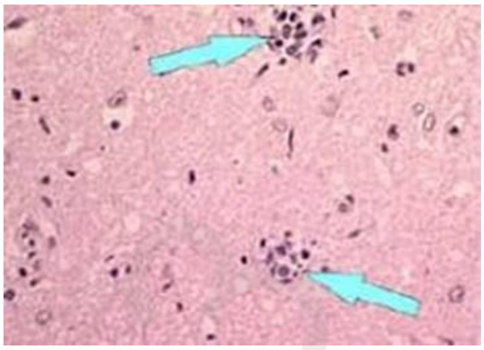
(4) virus antigen detection in biopsy specimen from corneal scrapings or skin from nape of neck.

(5) intracerebral inoculation of suckling mice.

(6) Detection of viral N.A (Nucleic Acid) by PCR.

Although each of the first 3 methods have distinct advantages, none provide a rapid definitive diagnosis.





Negri bodies (Rabies)

PREVENTION AND TREATMENT:

• No specific treatment

• Washing of wound with soap and water for 10 to 15 minutes.

•contacting a healthcare provider to determine if post-exposure prophylaxis is required.

• Vaccination: post-exposure and also pre-exposure for high-risk groups, e.g., veterinarians and animal handlers. (Rabies vaccine is 100% effective if given early, and still has a chance of success if delivery is delayed).

•Administration, post-exposure, of immunoglobulins to non-vaccinated persons.

Vaccination after exposure, post-exposure prophylaxis (PEP), is highly successful in preventing rabies. In unvaccinated humans, rabies is virtually always fatal after neurological symptoms have developed.

LECTURE 13-

Coronavirus

General Properties and Diseases: -

-A group of related RNA viruses that cause diseases in mammals and birds. In humans and birds, they cause respiratory tract infections that can range from mild to lethal. Mild illnesses in humans include some cases of the common cold (which is also caused by other viruses, predominantly rhinoviruses).

-More lethal varieties of the virus can cause SARS, MERS and COVID-19, which is causing the ongoing pandemic, in cows and pigs, they cause diarrhea, while in mice they cause hepatitis and encephalomyelitis.

-Coronaviruses are enveloped viruses with a positive-sense single-stranded RNA genome and a nucleocapsid of helical symmetry.

-The genome size of coronaviruses ranges from approximately 26 to 32 kilobases, one of the largest among RNA viruses.

- They have characteristic club-shaped spikes that project from their surface, which in electron micrographs create an image reminiscent of the stellar corona, from which their name derives.

- Coronaviruses are large, roughly spherical particles with unique surface projections.Their size is highly variable with average diameters of 80 to 120 nm. Extreme sizes are known from 50 to 200 nm in diameter.

-They are enclosed in an envelope embedded with a number of protein molecules. The lipid bilayer envelope, membrane proteins, and nucleocapsid protect the virus when it is outside the host cell.

-The membrane (M), envelope (E) and spike (S) structural proteins are anchored in the viral envelope.

- The E and M protein are the structural proteins that combined with the lipid bilayer to shape the viral envelope and maintain its size.

- S proteins are needed for interaction of the virus (receptor binding and membrane fusion) with the host cells. But human coronavirus NL63 is peculiar in that its M protein has the binding site for the host cell, and not its S protein.

- The spikes are the most distinguishing feature of coronaviruses and are responsible for the corona- or halo-like surface. On average a coronavirus particle has 74 surface spikes. Each spike is about 20 nm long and is composed of a trimer of the S protein.

- S1 proteins are the most critical components in terms of infection. They are also the most variable components as they are responsible for host cell specificity.

- A subset of coronaviruses (specifically the members of betacoronavirus subgroup A) also has a shorter spike-like surface protein called hemagglutinin esterase (HE) which help in the attachment to and detachment from the host cell.

-Inside the envelope, there is the nucleocapsid, which is formed from multiple copies of the nucleocapsid (N) protein, which are bound to the positive-sense single-stranded RNA genome in a continuous Beads-on-a-string type conformation.

Transmission: -

Infected carriers are able to shed viruses into the environment. The interaction of the coronavirus spike protein with its complementary cell receptor is central in determining the tissue tropism, infectivity, and species range of the released virus.

Coronaviruses mainly target epithelial cells. They are transmitted from one host to another host, depending on the coronavirus species, by either an aerosol, fomite, or fecal-oral route.

Human coronaviruses infect the epithelial cells of the respiratory tract, while animal coronaviruses generally infect the epithelial cells of the digestive tract. SARS coronavirus, for example, infects the human epithelial cells of the lungs via an aerosol route by binding to the angiotensin-converting enzyme 2 (ACE2) receptor.

Transmissible gastroenteritis coronavirus (TGEV) infects the pig epithelial cells of the digestive tract via a fecal-oral route.

Classification of Coronaviruses: -

Coronaviruses form the subfamily Orthocoronavirinae which is one of two sub-families in the family Coronaviridae, order Nidovirales, and realm Riboviria.

They are divided into the four genera:

1. Alphacoronavirus,
2. Betacoronavirus,
3. Gammacoronavirus,
4. Deltacoronavirus.

Alphacoronaviruses and betacoronaviruses infect mammals, while gammacoronaviruses and deltacoronaviruses primarily infect birds.

Origin of Coronaviruses: -

Origins of human coronaviruses with possible intermediate hosts: -

Bats and birds, as warm-blooded flying vertebrates, are an ideal natural reservoir for the coronavirus gene pool (with bats the reservoir for alphacoronaviruses and betacoronavirus – and birds the reservoir for gammacoronaviruses and deltacoronaviruses). The large number and global range of bat and avian species that host viruses have enabled extensive evolution and dissemination of coronaviruses.

The human coronavirus NL63 shared a common ancestor with a bat coronavirus (ARCoV.2) between 1190 and 1449 CE. More recently, alpaca coronavirus and human coronavirus 229E diverged sometime before 1960.

MERS-CoV emerged in humans from bats through the intermediate host of camels. MERS-CoV, although related to several bat coronavirus species, appears to have diverged from these several centuries ago. The most closely related bat coronavirus and SARS-CoV diverged in 1986. The ancestors of SARS-CoV first infected leaf-nose bats of the genus; subsequently, they spread to horseshoe bats in the species Rhinolophidae, then to Asian palm civets, and finally to humans.

Unlike other betacoronaviruses, bovine coronavirus of the species Betacoronavirus 1 and subgenus Embecovirus is thought to have originated in rodents and not in bats.In the 1790s, equine coronavirus diverged from the bovine coronavirus after a cross-species jump. Later in the 1890s, human coronavirus OC43 diverged from bovine coronavirus after another cross-species spillover event. It is speculated that the flu pandemic of 1890 may have been caused by this spillover event, and not by the influenza virus, because of the related timing, neurological symptoms, and unknown causative agent of the pandemic.

Besides causing respiratory infections, human coronavirus OC43 is also suspected of playing a role in neurological diseases. In the 1950s, the human coronavirus OC43 began to diverge into its present genotypes.

Phylogenetically, mouse hepatitis virus (Murine coronavirus), which infects the mouse's liver and central nervous system, is related to human coronavirus OC43 and bovine coronavirus. Human coronavirus HKU1, like the aforementioned viruses, also has its origins in rodents.

Infection in humans: -

Coronaviruses vary significantly in risk factor. Some can kill more than 30% of those infected, such as MERS-CoV, and some are relatively harmless, such as the common cold.

Coronaviruses can cause colds with major symptoms, such as fever, and a sore throat from swollen adenoids. Coronaviruses can cause pneumonia (either direct viral pneumonia or secondary bacterial pneumonia) and bronchitis (either direct viral bronchitis or secondary bacterial bronchitis).

The human coronavirus discovered in 2003, SARS-CoV, which causes severe acute respiratory syndrome (SARS), has a unique pathogenesis because it causes both upper and lower respiratory tract infections.

Six species of human coronaviruses are known, with one species subdivided into two different strains, making seven strains of human coronaviruses altogether.

Four human coronaviruses produce symptoms that are generally mild, even though it is contended they might have been more aggressive in the past.

1. Human coronavirus OC43 (HCoV-OC43), β-CoV

2. Human coronavirus HKU1 (HCoV-HKU1), β-CoV

3. Human coronavirus 229E (HCoV-229E), α-CoV

4. Human coronavirus NL63 (HCoV-NL63), α-CoV–

Three human coronaviruses produce potentially severe symptoms:

1. severe acute respiratory syndrome coronavirus (SARS-CoV), β-CoV (identified in 2003).

2. Middle East respiratory syndrome-related coronavirus (MERS-CoV), β-CoV (identified in 2012).

3. severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), β-CoV (identified in 2019).

These cause the diseases commonly called SARS, MERS, and COVID-19 respectively.

Common cold: -

Although the common cold is usually caused by rhinoviruses, in about 15% of cases the cause is a coronavirus. The human coronaviruses HCoV-OC43, HCoV-HKU1, HCoV-229E, and HCoV-NL63 continually circulate in the human population in adults and children worldwide and produce the generally mild symptoms of the common cold.] The four mild coronaviruses have a seasonal incidence occurring in the winter months in temperate climates. There is no preponderance in any season in tropical climates.

What are the stages of coronavirus infection?

There are three general phases of infection with SARS-Cov-2, the coronavirus that causes COVID-19: -

1. Incubation period: This is the time between getting infected and when symptoms appear. In general, you may see symptoms start two to 14 days after infection. The incubation period varies among individuals, and it varies depending on the variant. Even though you do not have symptoms in the incubation period, you can transmit the coronavirus to another person during this stage.

This is why, if you suspect you were exposed to someone with COVID-19, you should self-quarantine, watch for symptoms and consider getting tested four or five days following the exposure. This way, you can help prevent the spread of COVID-19. Please review Centers for Disease Control and Prevention (CDC) guidelines for isolation and quarantine.

2.Acute COVID-19: Once symptoms appear, you have entered the acute stage. You may have fever, cough and other COVID-19 symptoms. Active illness can last one to two weeks if you have mild or moderate coronavirus disease, but severe cases can last months. Some people are asymptomatic, meaning they never have symptoms but do have COVID-19.

If you develop symptoms or suspect you are asymptomatically infected, call your health care provider, follow testing guidelines, and follow all isolation and safety guidelines.

3.COVID-19 recovery: Post-COVID-19 symptoms, such as lingering cough, on and off fever, weakness, and changes to your senses of smell or taste, can persist for weeks or even months after you recover from acute illness. Persistent symptoms are sometimes known as long COVID-19.

Prevention and treatment: -

A number of vaccines using different methods have been developed against human coronavirus SARS-CoV-2. Antiviral targets against human coronaviruses have also been identified such as viral proteases, polymerases, and entry proteins. Drugs are in development which target these proteins and the different steps of viral replication.

Vaccines are available for animal coronaviruses although their effectiveness is limited. In the case of outbreaks of highly contagious animal coronaviruses, such as PEDV, measures such as destruction of entire herds of pigs may be used to prevent transmission to other herds.

Laboratory Diagnosis of Coronaviruses: -

In order to stop the spread, researchers worldwide are working around the clock aiming to develop reliable tools for early diagnosis of severe acute respiratory syndrome (SARS-CoV-2) understanding the infection path and mechanisms. Currently,

1. nucleic acid-based molecular diagnosis (real-time reverse transcription polymerase chain reaction (RT-PCR) test) is considered the gold standard for early diagnosis of SARS-CoV-2.

2. Antibody-based serology detection is ineffective for the purpose of early diagnosis, but a potential tool for serosurveys, providing people with immune certificates for clearance from COVID-19 infection.
3. Methods used for diagnosis of the discovered human coronavirus (SARS, MERS, COVID-19) include:
   - nucleic acid detection,
   - gene sequencing,
   - antibody detection,
   - antigen detection, and,
   - clinical diagnosis.

   Test results may remain positive for weeks to several months following infection, but this does not necessarily mean you are still infectious. Most people are no longer infectious beyond the recommended isolation precautions period.

If the infected person has conditions that cause severe immunosuppression, contact your health care provider to determine how long he should be isolated and how to determine when he is no longer potentially infectious to others.

LECTURE- 14

Bacteriophages (Bacterial viruses)

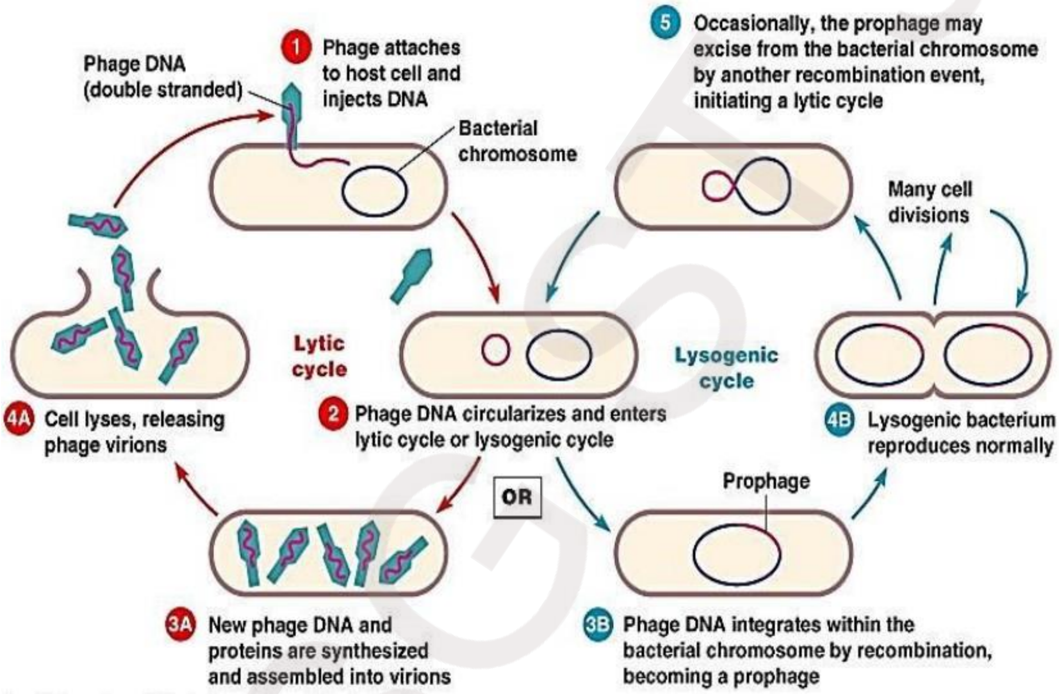
Bacteriophages are viruses that infect bacteria. Replicating within the bacterial cell therefore they are obligate parasites. Infection with bacteriophages is restricted to particular strains within a single bacterial species. Phage exist in many forms and infect all living systems such as animals ,plants, insects and bacteria therefore phages are ubiquitous in nature and they have been showed to be found in soil and sediment. The death of the host cell result from the release of the progeny and replication of viral particles. About 20-40% of marine bacteria every day have been killed by bacteriophages. Therefore, they play important role in bacterial evolution and ecological systems, and have a considerable role in biogeochemical cycles (carbon, nitrogen, sulfur and phosphorous cycles).

Composition:

Depending upon the phage, the nucleic acid can be either DNA or RNA but not both. The nucleic acids of phages often contain unusual or modified bases, which protect phage nucleic acid from nucleases that break down host nucleic acids during phage infection. Simple phages may have only 3-5 genes while complex phages may have over 100 genes. The phages majority contain double strand DNA (dsDNA), while there are small phage groups with ssRNA, dsRNA, or ssDNA. There are three morphological forms of phages: filamentous phages, isosahedral phages without tails, phages with tails, and even several phages with a lipid-containing envelope or contain lipids in the particle shell.

Bacteriophage Replication Cycle and Classification of Bacteriophages:

The phage infection cycle is important when choosing a phage for antibacterial application. All known bacteriophages can be divided into two groups according to the type of infection. One group is characterized by a lytic infection and the other is represented by a lysogenic, or temperate. In the first form of infection (lytic cycle) occurs once when a host cell infected by virulent phage immediately begins to exploit the metabolic machinery of the cell and directs it towards replication of new virion particles in which the new phage DNA has been packaged and the protein capsid is fully formed phage-encoded proteins. holins and endolysins work together to cause lysis of the cell causing death of the host bacterial cell and the progeny are released. As aresult, new phages are released into the extracellular space. The other mode of infection, a temperate phage has the ability to enter a lysogenic cycle, in which the phage DNA is integrated into the host genome. The DNA is replicated along with the host genome. Such transition of viral DNA could take place through several generations of bacterium without major metabolic consequences for it. Eventually the phage genes, at certain conditions impeding the bacterium state, will revert to the lytic cycle, leading to release of fully assembled phages.



Types of viral life cycles

The bacteriophages classification is depended on several factors such as their host preference, viral morphology, genome type, and auxiliary structures such as tails or envelopes. The bacteriophages.The phage morphology and nucleic acid properties are key classification factors. The phages majority contain double strand DNA (dsDNA), while there are small phage groups with ssRNA, dsRNA, or ssDNA (ss stands for single strand). There are a few morphological groups of phages: filamentous phages, isosahedral phages without tails, phages with tails, and even several phages with a lipid-containing envelope or contain lipids in the particle shell. This makes bacteriophages the largest viral group in nature. More than 5500 bacterial viruses have been examined in the electron microscope.

Phage Therapy:

Phage therapy involves clinical treatment of bacterial infections with phages (bacteriophages). The method, which has gained a renewed interest because of increasing frequency of infections by multidrug-resistant bacteria, has potential benefits. Phages are highly effective in killing their targeted bacteria (their action is bactericidal). Phages may be considered as good alternative for patients allergic to antibiotics.

Phage therapy benefits

● Phages work against both treatable and antibiotic-resistant bacteria.

● They may be used alone or with antibiotics and other drugs.

● Phages multiply and increase in number by themselves during treatment (only one dose may be needed).

● They only slightly disturb normal “good” bacteria in the body.

● Phages are natural and easy to find.

● They are not harmful (toxic) to the body.

●They are not toxic to animals, plants, and the environment. Phage therapy disadvantages

● Phages are currently difficult to prepare for use in people and animals.

● It’s not known what dose or amount of phages should be used.

● It’s not known how long phage therapy may take to work.

● It may be difficult to find the exact phage needed to treat an infection.

● Phages may trigger the immune system to overreact or cause an imbalance.

● Some types of phages don’t work as well as other kinds to treat bacterial infections.

● There may not be enough kinds of phages to treat all bacterial infections.

● Some phages may cause bacteria to become resistant.

Questions

Q1:-Enumerate phage therapy benefits ? Q2:-Define Bacteriophages

LECTURE 15-

Oncogenic viruses and Antiviral Drugs &Viral vaccines

Oncogenic viruses (Human cancer viruses)

Introduction

Cell growth: is the cell proliferation (the increase in cell numbers that occurs through repeated cell division).

• Cell growth is regulated by two groups of regulatory genes:

A. Proto-oncogenes (cellular oncogene, c-onc) Are normal genes which control cell proliferation, but which have the potential to contribute to cancer development if their expression is altered (changed into oncogenes). So, Oncogenes are genes that cause cancer.

B- Tumor Suppressor Genes include genes that inhibit cell growth, fixing broken DNA or causing a cell to die. Examples: P53, Rb (retinoblastoma). In normal cells, oncogenes are ‘‘switched off’’or down-regulated by antioncogeneproteins. An oncovirus is a virus that can cause cancer.

• It refers to any virus with a DNA or RNA genome causing cancer and also called "tumor virus" or "cancer virus".

• Most viruses are non-transforming

- however, they may play a role in reducing the host cell’s ability to inhibit apoptosis.

Oncogenic virus: A virus that is able to cause cancer is known as an oncogenic virus. Evidence that a virus is oncogenic includes the regular presence in the tumor cells of virus DNA, which might be all or a part of the virus is possible that the virus is just one of a number of carcinogenic factors that can give rise to these cancers. At least 15-20% of all human tumors worldwide have a viral cause. The viruses that have been strongly associated with human cancers. Viruses are etiologic factors in the development of several types of human tumors, including two of great significance worldwide cervical cancer and liver cancer. They include human papillomaviruses (HPVs), Epstein-Barr virus (EBV), human herpesvirus 8, hepatitis B virus, hepatitis C virus, and two human retroviruses plus several candidate human cancer viruses. Many viruses can cause tumors in animals, either as a consequence of natural infection or after experimental inoculation.

CLASSES OF ONCOGENIC VIRUSES:

There are two classes of tumor viruses:

● DNA tumor viruses

● RNA tumor viruses, the latter also being referred to as Retroviruses.

DNA tumor viruses :-

1- Papovaviridae include human papilloma virus causes uterine (cervical) cancer.

2- polyomaviridae include JK, BK causes solid tumor in rodents and Merkel Cell Polyoma virus cause Merkel Cell Carcinoma (rare skin cancer).

3- Herpesviridae include: -

a) EBV infection increases the risk of Burkitt lymphoma, Nasopharyngeal carcinoma and some types of Hodgkin’s and non-Hodgkin’s lymphoma also stomach cancer.

b) Kaposi's sarcoma-associated herpesvirus (KSHV or HHV-8) is associated with Kaposi’s sarcoma, a type of skin cancer.

c) Human cytomegalovirus (CMV or HHV-5) is associated with mucoepidermoid carcinoma and possibly other malignancies.

4- Hepadnaviridae include Hepatitis B virus causes hepatocellular carcinoma.

5- Adenoviridae include adenovirus causes various solid tumor in rodents.

6- Poxviridae include Smallpox; cowpox causes various solid tumor.

RNA tumor Viruses

1-Retroviridae include Human T-cell leukemia virus (HTLV-1; HTLV-2) which causes Adult T-cell leukemia, Lymphoma.

2-Flaviviridae include Hepatitis C Virus causes Hepatocellular carcinoma.

General Features of Viral Carcinogenesis

1- Viruses can cause cancer in animals and humans.

2- Tumor viruses frequently establish persistent infections in natural host.

3- Viruses are seldom complete carcinogens.

4- Host factors are important determinants of virus-induced tumorigenesis

5. Virus infections are more common than virus-related tumor formation.

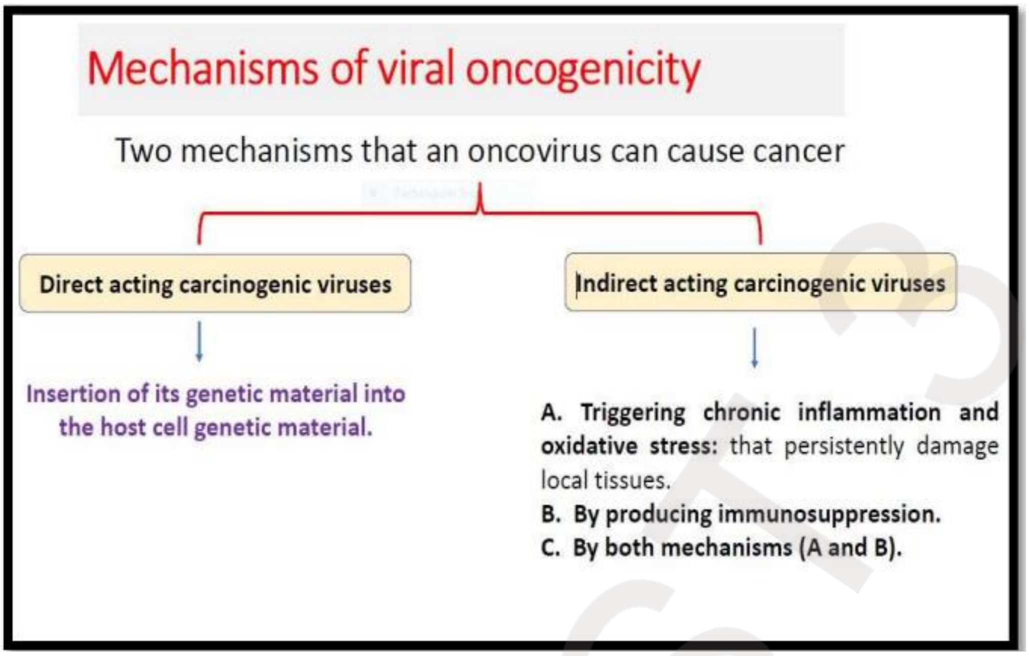
6. Long latent periods usually elapse between initial virus infection and tumor appearance.

7- Viral strains may differ in oncogenic potential.

8- Viruses may be either direct- or indirect-acting carcinogenic agents.

9- Oncogenic viruses modulate growth control pathways in cells.

10- Animal models may reveal mechanisms of viral carcinogenesis.



Anti-viral Drugs:

Antiviral drugs are a class of medication used specifically for treating viral infections. Like antibiotics for bacteria, specific antivirals are used for specific viruses. Designing safe and effective antiviral drugs is difficult, because viruses use the host's cells to replicate. This makes it difficult to find targets for the drug that would interfere with the virus without harming the host organism's cells. Moreover, the major difficulty in developing vaccines and anti-viral drugs is due to viral variation.

The number of antiviral drugs is very small because:

1. The virus is obligate intracellular parasite, difficulty in obtaining selective toxicity against virus.

2. Relatively ineffective, because many cycle of viral patients is well. by the time the patients have systemic viral disease .

3. Some virus remain latent in cell e.g. Herpes virus family

4. The emergence of viral drug resistance viral mutates.

Antiviral drugs are available to treat only a few viral diseases.

The reason for this is the fact that viral replication is so intimately associated with the host cell that any drug that interferes significantly with viral replication, is likely to be toxic to the host. Most of the antiviral drugs now available are designed to help deal with HIV, herpes viruses, the hepatitis B and C viruses, and influenza A and B viruses. Researchers are working to extend the range of antivirals to other families of pathogens. Two useful antiviral are: One way of doing this is to develop nucleotide or nucleoside analogues (Nucleotide analogues:These are synthetic compounds which resemble nucleosides , but have an incomplete or abnormal deoxy-ribose /or ribose group) that look like the building blocks of RNA or DNA, but deactivate the enzymes that synthesize the RNA or DNA once the analogue is incorporated. This approach is more commonly associated with the inhibition of reverse transcriptase (RNA to DNA)

.

Stages in virus replication which are possible targets for chemotherapeutic agents:

- Attachment to host cell
- Uncoating - (Amantadine)
- Synthesis of viral mRNA
- Translation of mRNA (Interferon)
- Replication of viral RNA or DNA (Interferon)
- Maturation of new virus proteins (Protease inhibitors)
- Assembly, release :- Protease inhibitors can be developed to prevent the final maturation of viral proteins in viruses that use a polyprotein expression strategy

Rifampicin and Tamiflu. vaccine:- is a biological preparation that provides active acquired immunity to a particular disease. A vaccine typically contains an agent that resembles a diseasecausing micro-organism and is often made from weakened or killed forms of the microbe, its toxins or one of its surface proteins.

Vaccines can be prophylactic or therapeutic (e.g., vaccines against cancer). Vaccines are very effective on stable viruses, but are of limited use in treating a patient who has already been infected. They are also difficult to successfully deploy against rapidly mutating viruses, such as influenza (the vaccine for which is updated every year) and HIV. Antiviral drugs are particularly useful in these cases.

Attributes of a good vaccine

1. Ability to elicit the appropriate immune response for the particular pathogen.
2. Long term protection.
3. Safety.
4. Stable.
5. Inexpensive.

Types of Vaccines

1- Live, attenuated vaccines

2- Inactivated vaccines (killed vaccine)

3- Subunit vaccines

4- Toxoid vaccines

5- DNA vaccines

6- Recombinant vector vaccines

- **Attenuated Live Vaccines**

vaccines contain live, attenuated microorganisms. Many of these are active viruses that have been cultivated under conditions that disable their virulent properties, and become less dangerous organisms to produce a broad immune response. Although most attenuated vaccines are viral, some are bacterial in nature. Examples include the viral diseases measles, rubella, and mumps, and the bacterial disease typhoid.

- **Killed Viral Vaccines**

Vaccines contain inactivated virus, but previously virulent, micro-organisms that have been destroyed with chemicals, heat, radiation, or antibiotics without destroying the antigenicity of the virus. Examples are influenza, cholera, hepatitis A, and rabies.

- **Subunit vaccines:**

viral proteins or groups of proteins are used. These proteins can be purified directly from viral particles. However this is expensive, since it is difficult to prepare virus in large enough quantities for protein purification, and potentially dangerous since there is the possibility of contaminating virulent virus.

- **DNA- Based Vaccines genes (DNA)**

encoding specific viral proteins are injected into an animal (either in muscle or skin). The DNA is then taken up by cells, where it is transcribed into mRNA which is then translated to give rise to the viral protein. This protein is expressed on the surface of cells, either alone or in association with MHC molecules. It is recognized as a foreign molecule by the immune system, and elicits an immune response.

- **Toxoid Vaccines:**

For bacteria that secrete toxins, or harmful chemicals. These vaccines are used when a bacterial toxin is the main cause of illness. they can inactivate toxins by treating them with formalin Such “detoxified” toxins, called toxoids, and are safe for use in vaccines.

- **Recombinant vector vaccines**

Immunogenic proteins of virulent organisms may be synthesized artificially by introducing the gene coding for the protein into an expression vector, such as E-coli or yeasts.

Q1:-Enumerate types of viral vaccine ?

Q2:-Define Antiviral drugs?